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From Conformal Blocks to Double-Layer Geometry: Monodromy, Spherical Quotients, and Modular Parameters

Yuanting Liu

Department of Physics and Astronomy,
Stony Brook University,
Stony Brook, NY 11794-3800, USA

E-mail: yuanting.liu@stonybrook.edu

Abstract: We study a geometric realization of two-dimensional unitary conformal field theories in which the analytic structure of conformal blocks is related to a global double-layer geometry. Starting from the BPZ equations of unitary Virasoro minimal models, we analyze the associated hypergeometric conformal blocks and their projective monodromy. The Euler-period representation naturally leads to a family of cyclic superelliptic curves whose geometry encodes the periods of the conformal blocks. We identify a canonical order-two subgroup of the cyclic deck transformation group and show that its quotient is a genus-zero curve,

$$C_p/\mathbb{Z}_2 \simeq \mathbb{P}^1 \simeq S^2. \tag{0.1}$$

This provides a spherical quotient geometry that can be related to a hyperelliptic double-layer realization, in which the $\mathbb{Z}_2$ involution exchanges the two sheets.

We then formulate the corresponding global geometry in terms of the distinction between the discrete layer-exchange symmetry and the continuous gluing direction. The former is described by the $\mathbb{Z}_2$ involution, while the latter is represented by a circle fiber, leading in the trivial case to

$$S^1 \hookrightarrow M \longrightarrow S^2, \qquad M \simeq S^2 \times S^1. \tag{0.2}$$

This geometry provides a natural setting for relating the double-layer construction to stable systolic geometry and for allowing modular data to vary continuously along the gluing direction. In particular, rather than assigning a single modular parameter to each layer, we associate τ_α with individual quasiparticle sectors and propose a modular diffusion functional describing their variation along S^1.

Finally, we consider $S^2 \times S^2$ as a pre-target geometry for the two-CFT system. A map to the two CFT layers admits a stable section, providing the geometric stability condition for the double-layer realization. For two Ising layers, the Weyl-anomaly contributions have the same sign and add, with $c_1 = c_2 = 1/2$ and $c_{\mathrm{dbl}} = 1$, giving a curvature response compatible with the positively curved product geometry. The anomaly and curvature analysis is used as a consistency check, while stability is associated with the stable section. Together, these constructions connect conformal-block monodromy, algebraic curves, hyperelliptic double layers, global systolic geometry, and sector-dependent modular data within a common geometric framework.

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1 Conformal Blocks and Modular Geometry

1.1 BPZ Equation and boundary correlation

The primary fields of two-dimensional conformal field theories, in particular the minimal models, are characterized by the representation theory of the Virasoro algebra. The allowed conformal weights are determined by the Kac determinant, which signal the appearance of null descendant states at specific levels. At these values, the corresponding Verma modules become reducible and contain proper submodules generated by the null states. The decoupling of these null descendants from physical correlation functions subsequently imposes differential constraints on correlation functions, leading to the BPZ differential equations. These equations were systematically studied by Belavin, Polyakov and Zamolochikov and provide one of the most powerful algebraic method for determining conformal correlation functions and conformal blocks.

Consider a highest weight representation of Virasoro algebra, denoted by $|h\rangle$,

$$L_0|h\rangle = h|h\rangle, \qquad L_n|h\rangle = 0 \quad n > 0 \tag{1.1}$$

At level n, the descendant states are

$$L_{-n_1}L_{-n_2}\cdots L_{-n_k}|h\rangle, \quad n_1 + n_2 + \cdots + n_k = n \tag{1.2}$$

The innner products of there descendant states form a Gram matrix, it's determinant is Kac determinant

$$\det M_n = \prod_{r,s>1 rs\leq 0} (h - h_{r,s})^{P(N-rs)} \tag{1.3}$$

where $P(n)$ is the number of integer partition of n. And, suppose now consider minimal model $\mathcal{M}(p, p')$

$$h_{r,s} = \frac{(p'r - ps)^2 - (p - p')^2}{4pp'} \quad h_{r,x} = h_{p'-r,p-s} \tag{1.4}$$

the numbder of distinct primary field is

$$N_{\text{primary}} = \frac{(p-1)(p'-1)}{2} \tag{1.5}$$

We consider the simplest case $\phi_{2,1}$'s null states at level two. The level-two descendants of the highest weight state $|h_{2,1}\rangle$ are generated by

$$L_{-2}|h_{2,1}\rangle, \quad L_{-1}^2|h_{2,1}\rangle \tag{1.6}$$

their linear combination should give a null vector

$$\left(L_{-1}^2 - \frac{2(2h_{2,1}+1)}{3}L_{-2}\right)|h_{2,1}\rangle = 0 \tag{1.7}$$

equivalently,

$$\left(\partial_z^2 - \frac{2(2h_{2,1}+1)}{3}L_{-2}\right)\phi_{2,1}(z) = 0 \tag{1.8}$$

After extracting the appropriate conformal prefactor, the holomorphic four-point conformal block $\mathcal{F}(z)$ satisfies a second-order Fuchsian differential equation of the form

$$\left(\partial_z^2 + \left(\frac{A}{z} + \frac{B}{z-1}\right)\partial_z + \frac{C}{z^2} + \frac{D}{(z-1)^2} + \frac{E}{z(z-1)}\right)\mathcal{F}(z) = 0 \tag{1.9}$$

The coefficients A, B, C, D, E are determined by the conformal weights of the external fields and by the central charge. With an appropriate extraction of the kinematic prefactor, this equation becomes the Gauss hypergeometric equation,

$$z(1-z)\mathcal{F}''(z) + \big[c_{\text{hyp}} - (a+b+1)z\big]\mathcal{F}'(z) - ab\,\mathcal{F}(z) = 0 \tag{1.10}$$

In general, the solutiones are Hypergeometric function,

$${}_2F_1(a,b;c;z) = \sum_{n=0}^{\infty} \frac{(a,n)(b,n)}{(c,n)}\frac{z^n}{n!} \tag{1.11}$$

where

$$(a,0)=1 \quad (a,n)=\prod_{i=0}^{n-1}(a+i) \tag{1.12}$$

the coefficient

$$\sum_{n=0}^{\infty}\frac{(a,n)(b,n)}{(c,n)n!}=\frac{\Gamma(a+n)}{\Gamma(a)}\frac{\Gamma(b+n)}{\Gamma(b)}\frac{\Gamma(c)}{\Gamma(c+n)}\frac{\Gamma(1)}{\Gamma(n+1)} \tag{1.13}$$

$$=\frac{\Gamma(c)\Gamma(1)}{\Gamma(a)\Gamma(b)}n^{a+b-c-1}[1+\mathcal{O}(n^{-1})] \tag{1.14}$$

Four point conformal block satisfies

$$F(z)=z^{\alpha}(1-z)^{\beta}G(z) \tag{1.15}$$

the corresponding differential equation of $G(z)$ part is

$$z(1-z)G''+[c-(a+b+1)z]G'-abG=0 \tag{1.16}$$

the solutions are

$$G_1(z)={}_2F_1(a,b,c;z) \tag{1.17}$$

$$G_2(z)=z^{1-c}{}_2F_1(a-c+1,b-c+1,2-c;z) \tag{1.18}$$

define

$$q(z)=\frac{G_1(z)}{G_2(z)} \tag{1.19}$$

the Shwarzian derivative

$$\{q(z),z\}=\frac{q(z)'''}{q(z)'}-\frac{3}{2}(\frac{q(z)''}{q(z)'})^2=2Q(z) \tag{1.20}$$

It means that the BPZ equation defines a projective structure on the punctured sphere

$$\mathbb{P}^1\setminus\{0,1,\infty\} \tag{1.21}$$

the vector valued function under monodromy group

$$\begin{pmatrix}G_1(z)\\G_2(z)\end{pmatrix}\mapsto M\begin{pmatrix}G_1(z)\\G_2(z)\end{pmatrix} \quad M\in SL(2,\mathbb{C}) \tag{1.22}$$

$$q(z)\mapsto\frac{aq(z)+b}{cq(z)+d} \tag{1.23}$$

the Mobius transformation. near these three fix points

$$\text{Near } z=0 \quad q(z)\sim 1+z^{1-c} \tag{1.24}$$

$$\text{Near } z=1 \quad q(z)\sim 1+(1-z)^{c-a-b} \tag{1.25}$$

$$\text{Near } z=\infty \quad q(z)\sim z^{-a}+z^{-b} \tag{1.26}$$

$$\tag{1.27}$$

the exponent differences are

$$\lambda_0 = 1 - c \tag{1.28}$$

$$\lambda_1 = c - a - b \tag{1.29}$$

$$\lambda_\infty = a - b \tag{1.30}$$

1.2 Geometry and cellulation

Take a regular (p, q) tessellation that every face is a p-gon, and q faces meet at every vertex. Take a center O, a vertex V, and a midpoint M of one edge, it gives a right triangle $\triangle OVM$, the three angles are

$$A = \frac{\pi}{p} \quad B = \frac{\pi}{q} \quad C = \frac{\pi}{2} \tag{1.31}$$

these sides opposite the angles A, B, C are a, b, c. In particular, $c = l/2$, where l is the edge length of the polygon. For $C = \pi/2$, the hyperbolic cosine law becomes

$$\cosh c = \cosh a \cosh b \tag{1.32}$$

and

$$\sin A = \frac{\sinh a}{\sinh c} \quad \sin B = \frac{\sinh b}{\sinh c} \tag{1.33}$$

thus

$$\cosh c = \cosh \frac{l}{2} = \frac{\cos A \cos B}{\sin A \sin B} = \frac{\cos(\pi/p)}{\sin(\pi/q)} \tag{1.34}$$

For general triangle group $\triangle(p, q, r)$ with angles $\pi/p, \pi/q, \pi/r$. The hyperbolic cosine law gives

$$\cosh a = \frac{\cos(\pi/p) + \cos(\pi/q)\cos(\pi/r)}{\sin(\pi/q)\sin(\pi/r)} \tag{1.35}$$

$$\cosh b = \frac{\cos(\pi/q) + \cos(\pi/r)\cos(\pi/p)}{\sin(\pi/r)\sin(\pi/p)} \tag{1.36}$$

$$\cosh c = \frac{\cos(\pi/r) + \cos(\pi/p)\cos(\pi/q)}{\sin(\pi/q)\sin(\pi/q)} \tag{1.37}$$

The quotient of upper half plane

$$\mathbb{H}/\Gamma \tag{1.38}$$

with orbifold $X(p, q, r)$. It's hyperbolic metric satisfies $\triangle u = e^{2u}$, where

$$ds^2 = e^{2u}|dz|^2 \tag{1.39}$$

for

$$f : \mathbb{H} \to \mathbb{D} \tag{1.40}$$

satisfies

$$\{f, z\} = 2Q(z) \tag{1.41}$$

where $Q(z)$ has poles determined by the celluation parameters (p, q, r). For three singularities at $\{0, 1, \infty\}$

$$Q(z) = \frac{1 - \frac{1}{p^2}}{4z^2} + \frac{1 - \frac{1}{q^2}}{4(z-1)^2} + \frac{\frac{1}{p^2} + \frac{1}{q^2} - \frac{1}{r^2} - 1}{4z(z-1)} \tag{1.42}$$

Spherical triangle groups

$$\frac{1}{p} + \frac{1}{q} + \frac{1}{r} > 1 \tag{1.43}$$

for examples

$$(2, 2, n) \quad (2, 3, 3) \quad (2, 3, 4) \quad (2, 3, 5) \tag{1.44}$$

and

$$g = 0 \tag{1.45}$$

Euclidean triangle groups

$$\frac{1}{p} + \frac{1}{q} + \frac{1}{r} = 1 \tag{1.46}$$

for examples

$$(2, 3, 6) \quad (2, 4, 4) \quad (3, 3, 3) \tag{1.47}$$

and

$$g = 1 \tag{1.48}$$

Hyperbolic triangle groups

$$\frac{1}{p} + \frac{1}{q} + \frac{1}{r} < 1 \tag{1.49}$$

they corresponds to the Fuchsian triangle groups

Suppose

$$X = \mathbb{H}/\Gamma \tag{1.50}$$

where Γ is torsion-free of index N. The genus

$$g = 1 + \frac{\mu}{2} - \frac{c}{2} \tag{1.51}$$

more general

$$2g - 2 = \frac{\text{Area(X)}}{2\pi} - c \tag{1.52}$$

c is the number of cusps, Area(X) is the hyperbolic area. For instance the well-known $PSL_2(\mathbb{Z})$ is the $(2, 3, \infty)$ cellulation.

1.3 Unitary CFT

For unitray minimal model

$$\mathcal{M}(p, p+1), \quad p \geq 2 \tag{1.53}$$

with central charge

$$c = 1 - \frac{6}{p(p+1)} \tag{1.54}$$

the conformal weight of $\phi_{(2,1)}$ is

$$h_{\phi_{(2,1)}} = \frac{(p+2)^2-1}{4p(p+1)} = \frac{p+3}{4p} \tag{1.55}$$

where the fusion of it is

$$\phi_{(2,1)} \times \phi_{(2,1)} = 1 + \phi_{(3,1)} \tag{1.56}$$

the level two null vector has universal form

$$(L_{-2} - \frac{3}{2(2h_{\phi_{(2,1)}}+1)} L_{-1}^2)|\phi_{(2,1)}\rangle = (L_{-2} - \frac{p}{p+1} L_{-1}^2)|\phi_{(2,1)}\rangle = 0 \tag{1.57}$$

for correlator

$$\langle \phi_{(2,1)}(\infty)\phi_{(2,1)}(1)\phi_{(2,1)}(z)\phi_{(2,1)}(0)\rangle \tag{1.58}$$

take the solution in form

$$z^{-2h}(1-z)^{-2h}\mathcal{F} \tag{1.59}$$

$$[\frac{p+1}{p}\partial_z^2 + \sum_i (\frac{h_i}{(z-z_i)^2} + \frac{1}{z-z_i}\partial_{z_i})]G = 0 \tag{1.60}$$

after fixing

$$z(1-z)\mathcal{F}'' + [c-(a+b+1)]\mathcal{F}' - ab\mathcal{F} = 0 \tag{1.61}$$

where

$$a = -\frac{1}{p} \quad b = \frac{3}{p+1} \quad c = 1 + \frac{2}{p+1} \tag{1.62}$$

thus

$${}_2F_1(-\frac{1}{p}, \frac{3}{p+1}; 1+\frac{2}{p+1}; z) \qquad z^{-2/(p+1)}{}_2F_1(-\frac{3}{p+1}, \frac{1}{p}; 1-\frac{2}{p+1}; z) \tag{1.63}$$

and these two solution corresponding to the two fusion channel, if exist, where

$$\sim 1 \quad \sim z^{-2/(p+1)} \tag{1.64}$$

corresponding there leading powers.

The three local exponent differences at $z = 0, 1, \infty$ are

$$\lambda = 1 - c \quad \mu = c - a - b \quad \nu = a - b \tag{1.65}$$

for our case

$$(\lambda, \mu, \nu) = (\frac{2}{p+1}, \frac{1}{p(p+1)}, \frac{4p+1}{p(p+1)}) \tag{1.66}$$

The absolute exponent differences are

$$\lambda = \frac{2}{p+1}, \qquad \mu = \frac{1}{p(p+1)}, \qquad \nu = \frac{4p+1}{p(p+1)},$$

up to the choice of signs associated with the local exponents. Their sum is

$$\Sigma = \lambda + \mu + \nu = \frac{2(3p+1)}{p(p+1)}.$$

Consequently,

$$\Sigma > 1 \quad \text{for } p \leq 5, \qquad \Sigma < 1 \quad \text{for } p \geq 6.$$

Thus the associated triangle-type geometry changes from the spherical regime to the hyperbolic regime at $p = 6$:

$$\begin{cases} \Sigma > 1, & p \leq 5, \qquad \text{spherical}, \\ \Sigma < 1, & p \geq 6, \qquad \text{hyperbolic}. \end{cases}$$

There is no integer $p \geq 2$ for which $\Sigma = 1$. This criterion concerns the geometric type of the associated triangle data. It should not by itself be identified with finiteness of the projective hypergeometric monodromy. Finite projective monodromy requires the corresponding exponent differences to satisfy the additional arithmetic conditions appearing in the Schwarz classification. We therefore use Σ here only to distinguish the spherical and hyperbolic triangle regimes.

Example 1. The Ising model is the unitary minimal model $\mathcal{M}(3,4), c = \frac{1}{2}$. Its three primary fields are

$$\phi_{(1,1)} \Longleftrightarrow 1, \tag{1.67}$$

$$\phi_{(1,2)} \Longleftrightarrow \sigma, \tag{1.68}$$

$$\phi_{(2,1)} \Longleftrightarrow \varepsilon, \tag{1.69}$$

with conformal weights

$$h_1 = 0, \qquad h_\sigma = \frac{1}{16}, \qquad h_\varepsilon = \frac{1}{2}. \tag{1.70}$$

The fusion rules are

$$\sigma \times \sigma = 1 + \varepsilon, \ \sigma \times \varepsilon = \sigma, \ \varepsilon \times \varepsilon = 1. \tag{1.71}$$

The energy operator ε. For the primary field $\varepsilon = \phi_{(2,1)}$, the level-two BPZ null-vector relation is

$$\left(L_{-2} - \frac{3}{4} L_{-1}^2 \right) |\varepsilon\rangle = 0. \tag{1.72}$$

Consider the four-point function

$$C_{\varepsilon,\varepsilon,\varepsilon,\varepsilon}(z) = \langle \varepsilon(\infty)\varepsilon(1)\varepsilon(z)\varepsilon(0) \rangle \,. \tag{1.73}$$

After extracting the standard conformal prefactor, the reduced function satisfies

$$z(1-z)F'' + (1-2z)F' - \frac{1}{4}F = 0. \tag{1.74}$$

This is the Gauss hypergeometric equation with

$$a = b = \frac{1}{2}, \qquad c = 1.$$

A convenient basis of solutions is therefore

$$F_1(z) = {}_2F_1\left(\frac{1}{2},\frac{1}{2};1;z\right), \qquad F_2(z) = {}_2F_1\left(\frac{1}{2},\frac{1}{2};1;1-z\right). \tag{1.75}$$

Although the physical fusion rule $\varepsilon \times \varepsilon = 1$ contains only the identity conformal family, the BPZ differential equation is second order and consequently has two linearly independent solutions. The physical s-channel boundary condition selects the appropriate local solution, while the second solution is naturally interpreted as the second period of the associated elliptic curve.

The hypergeometric function is related to the complete elliptic integral of the first kind by

$${}_2F_1\left(\frac{1}{2},\frac{1}{2};1;z\right) = \frac{2}{\pi}K(z). \tag{1.76}$$

Introducing the modular parameter

$$\tau = i\frac{K(1-z)}{K(z)}, \tag{1.77}$$

one obtains the modular lambda function

$$z = \lambda(\tau) = \frac{\vartheta_2(\tau)^4}{\vartheta_3(\tau)^4}, \tag{1.78}$$

and hence

$$K(z) = \frac{\pi}{2}\vartheta_3(\tau)^2. \tag{1.79}$$

The corresponding elliptic curve is

$$E_z: \qquad y^2 = x(x-1)(x-z). \tag{1.80}$$

Thus the cross-ratio z of the four-point function is identified with the modular parameter of the elliptic double cover through $z = \lambda(\tau)$. Equivalently,

$$\mathbb{P}^1 \setminus \{0,1,\infty\} \simeq \mathbb{H}/\Gamma(2),$$

with $\lambda(\tau)$ providing the corresponding modular coordinate. The hypergeometric monodromy is therefore naturally related to the modular group, while the two independent solutions $K(z)$ and $K(1-z)$ provide the two periods of E_z.

It is important, however, that the elliptic/modular structure above should not be confused with the spherical triangle structure arising from the spin field σ.

The spin operator σ. For

$$\sigma = \phi_{(1,2)}, \qquad h_\sigma = \frac{1}{16},$$

the level-two null-vector relation is

$$\left(L_{-2} - \frac{4}{3}L_{-1}^2\right)|\sigma\rangle = 0. \tag{1.81}$$

Consider

$$C_{\sigma\sigma\sigma\sigma}(z) = \langle \sigma(\infty)\sigma(1)\sigma(z)\sigma(0)\rangle \,. \tag{1.82}$$

The corresponding BPZ equation is again second order, reflecting the two fusion channels in

$$\sigma \times \sigma = 1 + \varepsilon.$$

A convenient basis of chiral conformal blocks is

$$\mathcal{F}_1(z) = \frac{1}{[z(1-z)]^{1/8}} \sqrt{\frac{1+\sqrt{1-z}}{2}} \tag{1.83}$$

$$\mathcal{F}_\varepsilon(z) = \frac{1}{[z(1-z)]^{1/8}} \sqrt{\frac{1-\sqrt{1-z}}{2}}. \tag{1.84}$$

Near $z = 0$, these behave as

$$\mathcal{F}_1(z) \sim z^{-1/8}, \quad \mathcal{F}_\varepsilon(z) \sim z^{3/8}. \tag{1.85}$$

The two leading powers differ by $\frac{3}{8} - \left(-\frac{1}{8}\right) = \frac{1}{2}$, which reflects the two possible intermediate conformal weights $h_1 = 0, h_\varepsilon = \frac{1}{2}$.

By crossing symmetry, the corresponding exponent difference is also $1/2$ at $z = 1$ and at $z = \infty$. Hence the projective exponent differences of the σ BPZ system are

$$(\lambda_0, \lambda_1, \lambda_\infty) = \left(\frac{1}{2}, \frac{1}{2}, \frac{1}{2}\right). \tag{1.86}$$

Their sum is

$$\Sigma = \lambda_0 + \lambda_1 + \lambda_\infty = \frac{3}{2} > 1, \tag{1.87}$$

so the associated triangle geometry is spherical. More precisely, the three exponent differences correspond to the triangle type $(2, 2, 2)$, namely the spherical triangle with three right angles. Its orientation-preserving subgroup is the Klein four group V_4.

Thus the σ conformal blocks provide a finite spherical projective realization, in contrast to the elliptic-period realization associated with the ε sector.

The hypergeometric function

$${}_2F_1\left(-\frac{1}{2}, \frac{1}{2}; 1; z\right) \tag{1.88}$$

should not be identified directly with the two physical σ conformal blocks above. Instead,

$${}_2F_1\left(-\frac{1}{2}, \frac{1}{2}; 1; z\right) = \frac{2}{\pi} E(z), \tag{1.89}$$

where $E(z)$ is the complete elliptic integral of the second kind. The modular parameter relevant to the elliptic curve is therefore naturally obtained from the ε sector through the first-kind integral $K(z)$, whereas the σ sector exhibits the finite spherical V_4 monodromy.

The Ising model consequently provides two complementary geometric realizations of the same four-point CFT data:

$$\sigma \longrightarrow \text{spherical } (2,2,2) \text{ geometry} \longrightarrow V_4 \tag{1.90}$$

$$\varepsilon \longrightarrow {}_2F_1\left(\frac{1}{2},\frac{1}{2};1;z\right) \longrightarrow K(z), K(1-z) \longrightarrow E_z \longrightarrow \lambda(\tau). \tag{1.91}$$

This distinction will be useful below when relating the spherical double-layer construction to the modular data carried by the individual conformal sectors.

Now consider the projective monodromy order

$$m_0 = \frac{p+1}{\gcd(2,p+1)} \quad m_1 = p(p+1) \quad m_\infty = \frac{p(p+1)}{\gcd(4p+1, p(p+1))} \tag{1.92}$$

simplify

$$(m_0, m_1, m_\infty) = (\frac{p+1}{\gcd(2,p+1)}, p(p+1), \frac{p(p+1)}{\gcd(3,p+1)}) \tag{1.93}$$

1.4 Characters of Minimal models

Recall the irreducible characters of minimal models:

$$\chi_{(r,s)}(q) = K_{r,s}^{(p,p')}(q) - K_{r,-s}^{(p,p')}(q) \tag{1.94}$$

where

$$K_{r,s}^{(p,p')}(q) = \frac{q^{-1/24}}{\varphi(q)} \sum_{n\in\mathbb{Z}} q^{(2pp'n+pr-p's)^2/4pp'} \tag{1.95}$$

take

$$q = ^{2\pi i\tau} \qquad \Lambda_\tau = \mathbb{Z} + \tau\mathbb{Z} \tag{1.96}$$

then define

$$\lambda_{r,s} = pr - p's \qquad \lambda'_{r,s} = pr + p's \tag{1.97}$$

then the character would be

$$\chi_{r,s}(\tau) = \frac{\Theta_{\lambda_{r,s}}^{(p,p')}(\tau) - \Theta_{\lambda'_{r,s}}^{(p,p')}(\tau)}{\eta(\tau)} \tag{1.98}$$

where

$$\Theta_\lambda^{(p,p')}(\tau) = \sum_{m\in\mathbb{Z}} q^{(2pp'm+\lambda)^2/4pp'} \tag{1.99}$$

$$= \sum_{m\in\mathbb{Z}} \exp[2\pi i\tau \frac{(2pp'm+\lambda)^2}{4pp'}] \tag{1.100}$$

thus

$$\Theta_\lambda^{(p,p')}(\tau) = \vartheta_{\frac{\lambda}{2pp'},0}(0, 2pp'\tau) \tag{1.101}$$

We introduce a vector valued modular forms

$$\Theta^{(p,p')}(\tau) = \begin{pmatrix} \vartheta_{0,0}(0, 2pp'\tau) \\ \vartheta_{\frac{1}{2pp'},0}(0, 2pp'\tau) \\ \cdots \quad \cdots \\ \vartheta_{\frac{2pp'-2}{2pp'},0}(0, 2pp'\tau) \\ \vartheta_{\frac{2pp'-1}{2pp'},0}(0, 2pp'\tau) \end{pmatrix} \tag{1.102}$$

in another words, the finite dimensional vector of all irreducible characters transforms invariant under S and T.

Recall fermionic free field realization in $\mathcal{N} = 1$ after bosonization, where

Sectors	Partition function (theta functions part)
R-R	$\theta_1 = \vartheta_{[0,0]}$
R-NS	$\theta_2 = \vartheta_{[\frac{1}{2},0]}$
NS-R	$\theta_4 = \vartheta_{[0,\frac{1}{2}]}$
NS-NS	$\theta_3 = \vartheta_{[\frac{1}{2},\frac{1}{2}]}$

Table 1. The theta function part of partition function of Non-linear sigama model after bosonization

$$\vartheta_{0,0}(z,\tau) = \sum_{n\in\mathbb{Z}} \exp(\pi i n^2\tau + 2\pi i n z) = \theta(z,\tau) \tag{1.103}$$

$$\vartheta_{0,\frac{1}{2}}(z,\tau) = \sum_{n\in\mathbb{Z}} \exp(\pi i n^2\tau + 2\pi i n(z+\frac{1}{2})) = \theta(z+\frac{1}{2},\tau) \tag{1.104}$$

$$\vartheta_{\frac{1}{2},0}(z,\tau) = \sum_{n\in\mathbb{Z}} \exp(\pi i (n+\frac{1}{2})^2\tau + 2\pi i (n+\frac{1}{2})z) = \exp(\frac{\pi i\tau}{4} + \pi i z)\theta(z+\frac{1}{2}\tau,\tau) \tag{1.105}$$

$$\vartheta_{\frac{1}{2},\frac{1}{2}}(z,\tau) = \sum_{n\in\mathbb{Z}} \exp(\pi i (n+\frac{1}{2})^2\tau + 2\pi i (n+\frac{1}{2})(z+\frac{1}{2})) = \exp(\frac{\pi i\tau}{4} + \pi i(z+\frac{1}{2}))\theta(z+\frac{1}{2}(1+\tau),\tau) \tag{1.106}$$

thus we could get minimal model by taking free fermionic realization in sector R-NS sector, but extend theta function and do extended by $z^{2pp'} - 1$ to vector valued theta function, in the end, taking the screening operators, you would get corresponding minimal model. This minimal model we get have another symmetry, which it is a $\mathcal{N} = 1$ supersymmetry minimal model.

1.5 Vector-valued modularity of conformal blocks

As discussed in the previous section, the conformal blocks of the minimal model $\mathcal{M}(3,4)$ associated with the primary field $\phi_{2,1}$ can be naturally expressed in terms of the elliptic integral $K(x)$. In this section, we investigate how this correspondence extends, at least, to the class of unitary minimal models.

For the unitary minimal model $\mathcal{M}(p, p+1)$, consider the hypergeometric equation associated with the four-point conformal blocks of the degenerate field $\phi_{2,1}$. We write two independent solutions as

$$ {}_2F_1^{(1)}(z), \qquad {}_2F_1^{(2)}(z), \tag{1.107} $$

and define the Schwarz, or uniformizing, coordinate by

$$ \tau_p(z) = \frac{{}_2F_1^{(2)}(z)}{{}_2F_1^{(1)}(z)}. \tag{1.108} $$

The associated hypergeometric equation gives the Schwarzian equation

$$ \tau_p, z = \frac{1-\alpha^2}{2z^2} + \frac{1-\beta^2}{2(1-z)^2} + \frac{1-\alpha^2-\beta^2+\gamma^2}{2z(1-z)}, \tag{1.109} $$

where

$$ \alpha = 1-c = -\frac{2}{p+1}, \qquad \beta = c-a-b = 1+\frac{1}{p(p+1)}, \qquad \gamma = a-b = -\frac{4p+1}{p(p+1)}. \tag{1.110} $$

Here

$$ a = -\frac{1}{p}, \qquad b = \frac{3}{p+1}, \qquad c = 1+\frac{2}{p+1}. \tag{1.111} $$

Under analytic continuation around the singular points $z = 0, 1, \infty$, the vector of solutions transforms by the corresponding monodromy matrices,

$$ \begin{pmatrix} {}_2F_1^{(1)} \\ {}_2F_1^{(2)} \end{pmatrix} \mapsto M_\gamma \begin{pmatrix} {}_2F_1^{(1)} \\ {}_2F_1^{(2)} \end{pmatrix} \tag{1.112} $$

The projective monodromy depends only on the exponent differences modulo integers. In particular,

$$ \alpha = -\frac{2}{p+1}, \qquad \beta = 1+\frac{1}{p(p+1)} \equiv \frac{1}{p(p+1)} \pmod{\mathbb{Z}}, \qquad \gamma = -\frac{4p+1}{p(p+1)}. \tag{1.113} $$

Thus the reduced denominator of the projective monodromy associated with γ is

$$ n_{\text{proj}} = \frac{p(p+1)}{\gcd(4p+1, p(p+1))} = \frac{p(p+1)}{\gcd(3, p+1)}. \tag{1.114} $$

Example 2. For the Ising model $\mathcal{M}(3,4)$, the four-punctured sphere

$$ \mathbb{P}^1 \setminus 0, z, 1, \infty \tag{1.115} $$

has a natural double cover given by the elliptic curve

$$ E_z : \qquad y^2 = x(x-1)(x-z). \tag{1.116} $$

The four branch points $0, z, 1, \infty$ on the sphere therefore define a genus-one double cover. Its modular parameter τ is related to the cross ratio z by the modular lambda function,

$$ z = \lambda(\tau) = \frac{\theta_2(\tau)^4}{\theta_3(\tau)^4}. \tag{1.117} $$

Using the Jacobi identity

$$\theta_3(\tau)^4 = \theta_2(\tau)^4 + \theta_4(\tau)^4, \tag{1.118}$$

one obtains

$$1 - z = \frac{\theta_4(\tau)^4}{\theta_3(\tau)^4}. \tag{1.119}$$

The two chiral conformal blocks of the Ising spin field may be written as

$$\mathcal{F}_{\pm}(z) = [z(1-z)]^{-1/8}\sqrt{\frac{1 \pm \sqrt{1-z}}{2}}. \tag{1.120}$$

Thus the hypergeometric conformal blocks on the four-punctured sphere admit an elliptic interpretation after passing to the double cover. In particular, the same cross-ratio z that parametrizes the punctured sphere becomes the modular lambda coordinate of the elliptic curve.

The four-punctured sphere also admits three inequivalent OPE channels, corresponding to the three pairings of the four marked points. We denote them by In the elliptic

Channels	Points	Characteristic functions
s	$(0, z)$	θ_3
t	$(z, 1)$	θ_4
u	$(0, 1)$	θ_3

Table 2. The three OPE channels of the four-punctured sphere.

description, these different pairings are related by the modular transformations of the elliptic curve.

We now consider the general unitary minimal model $\mathcal{M}(p, p+1)$. The hypergeometric block associated with $\phi_{2,1}$ can be written as

$${}_2F_1\left(-\frac{1}{p}, \frac{3}{p+1}; 1 + \frac{2}{p+1}; z\right) = {}_2F_1(a, b; c; z). \tag{1.121}$$

The standard Euler integral representation is

$${}_2F_1(a, b; c; z) = \frac{\Gamma(c)}{\Gamma(b)\Gamma(c-b)} \int_0^1 x^{b-1}(1-x)^{c-b-1}(1-zx)^{-a} dx. \tag{1.122}$$

Accordingly, we consider the differential

$$\omega_+ = x^{b-1}(1-x)^{c-b-1}(1-zx)^{-a}, dx. \tag{1.123}$$

For the parameters above,

$$b - 1 = -\frac{p-2}{p+1}, \qquad c - b - 1 = -\frac{1}{p+1}, \qquad -a = \frac{1}{p}. \tag{1.124}$$

To realize this differential on a cyclic cover, take

$$N = p(p+1), \tag{1.125}$$

and define

$$i = N(1-b) = p(p-2), \qquad j = N(1+b-c) = p, \qquad k = Na = -(p+1). \tag{1.126}$$

The corresponding superelliptic curve is

$$C_p(z): \qquad y^{p(p+1)} = x^{p(p-2)}(1-x)^p(1-zx)^{-(p+1)}. \tag{1.127}$$

On this curve,

$$\omega_+ = \frac{dx}{y}. \tag{1.128}$$

The corresponding period can therefore be written as

$$\Pi_1(z) = \int_\gamma \omega_+ = B\left(\frac{3}{p+1}, \frac{p}{p+1}\right) {}_2F_1\left(-\frac{1}{p}, \frac{3}{p+1}; 1+\frac{2}{p+1}; z\right), \tag{1.129}$$

up to the normalization determined by the choice of cycle γ.

The second local solution around $z = 0$ is obtained from the second Frobenius solution of the hypergeometric equation. It is therefore represented by a second period of the same family of cyclic covers,

$$\Pi_2(z) = \epsilon_p z^{-2/(p+1)} {}_2F_1\left(-\frac{3p+1}{p(p+1)}, \frac{1}{p+1}; 1-\frac{2}{p+1}; z\right), \tag{1.130}$$

where ϵ_p depends on the choice of branch and normalization of the corresponding Pochhammer cycle.

The four local monodromy exponents of the cyclic cover may be chosen as

$$\alpha_0 = p(p-2), \qquad \alpha_1 = p, \qquad \alpha_z = p+1, \qquad \alpha_\infty = p^2 - 2p - 1. \tag{1.131}$$

The corresponding ramification indices are

$$e'_s = \frac{N}{\gcd(N, \alpha_s)}, \qquad s = 0, 1, z, \infty. \tag{1.132}$$

The genus of the cyclic cover is consequently

$$g(C_p) = 1 - N + \frac{1}{2} \sum_{s=0,1,z,\infty} \left[N - \gcd(N, \alpha_s)\right]. \tag{1.133}$$

For $3 \le p \le 20$, this gives the following branching data:

Here e_s denotes the local monodromy exponent in the chosen Kummer convention, while e'_s denotes the corresponding ramification index. Explicitly, the exponents used in the eigenspace calculation are

$$(e_0, e_1, e_z, e_\infty) = \left(p(p-2), , p, , p^2-1, , 3p+1\right). \tag{1.134}$$

Taking

$$\sigma(x, y) = (x, \zeta_N^{-1} y), \tag{1.135}$$

p	cyclic deg N	$(e_0, e_1, e_z, e_\infty)$	$(e'_0, e'_1, e'_z, e'_\infty)$	$g(C_p)$	Projective order
3	12	$(3, 3, 8, 10)$	$(4, 4, 3, 6)$	7	$(2, 12, 12)$
4	20	$(8, 4, 15, 13)$	$(5, 5, 4, 20)$	14	$(5, 20, 20)$
5	30	$(15, 5, 24, 16)$	$(2, 6, 5, 15)$	17	$(3, 30, 10)$
6	42	$(24, 6, 35, 19)$	$(7, 7, 6, 42)$	33	$(7, 42, 42)$
7	56	$(35, 7, 48, 22)$	$(8, 8, 7, 28)$	45	$(4, 56, 56)$
8	72	$(48, 8, 63, 25)$	$(9, 9, 8, 72)$	52	$(9, 72, 24)$
9	90	$(63, 9, 80, 28)$	$(10, 10, 9, 45)$	76	$(5, 90, 90)$
10	110	$(80, 10, 99, 31)$	$(11, 11, 10, 110)$	95	$(11, 110, 110)$
11	132	$(99, 11, 120, 34)$	$(12, 12, 11, 66)$	104	$(6, 132, 44)$
12	156	$(120, 12, 143, 37)$	$(13, 13, 12, 156)$	138	$(13, 156, 156)$
13	182	$(143, 13, 168, 40)$	$(14, 14, 13, 91)$	162	$(7, 182, 182)$
14	210	$(168, 14, 195, 43)$	$(15, 15, 14, 210)$	175	$(15, 210, 70)$
15	240	$(195, 15, 224, 46)$	$(16, 16, 15, 120)$	217	$(8, 240, 240)$
16	272	$(224, 16, 255, 49)$	$(17, 17, 16, 272)$	248	$(17, 272, 272)$
17	306	$(255, 17, 288, 52)$	$(18, 18, 17, 153)$	263	$(9, 306, 102)$
18	342	$(288, 18, 323, 55)$	$(19, 19, 18, 342)$	315	$(19, 342, 342)$
19	380	$(323, 19, 360, 58)$	$(20, 20, 19, 190)$	351	$(10, 380, 380)$
20	420	$(360, 20, 399, 61)$	$(21, 21, 20, 420)$	370	$(21, 420, 140)$

Table 3. Branching data, genera, and projective orders for the cyclic covers C_p.

p	d_1	d_{N-1}	$\varphi(N)$	rational cyclotomic $h^{1,0}$	H^1 dimension
3	1	1	4	4	8
4	1	1	8	8	16
5	1	1	8	8	16
6	1	1	12	12	24
7	1	1	24	24	48
8	1	1	24	24	48
9	1	1	24	24	48
10	1	1	40	40	80
11	1	1	40	40	80
12	1	1	48	48	96
13	1	1	72	72	144
14	1	1	48	48	96
15	1	1	64	64	128
16	1	1	128	128	256
17	1	1	96	96	192
18	1	1	108	108	216
19	1	1	144	144	288
20	1	1	96	96	192

Table 4. Cyclotomic eigenspace data for the cyclic covers C_p.

the dimension of the σ-eigenspace indexed by r is given by the Chevalley–Weil formula

$$d_r = -1 + \sum_{s=0,1,z,\infty} \left\langle \frac{re_s}{N} \right\rangle , \tag{1.136}$$

where $\langle\cdot\rangle$ denotes the fractional part.

For the cases considered here, the eigenspaces $r = 1$ and $r = N-1$ each have dimension one. The corresponding cyclotomic contribution can be summarized as follows:

This construction exhibits a direct relation between the hypergeometric conformal blocks of the unitary minimal models and periods of a family of cyclic superelliptic curves. In the Ising case, the cyclic cover reduces to the familiar genus-one double-cover description and the period ratio becomes the modular parameter of the elliptic curve. For general p, the corresponding cyclic covers have higher genus, while the projective monodromy is still controlled by the exponent differences of the hypergeometric equation.

The cases $p = 3$ and $p = 5$ are distinguished by their familiar modular and statistical-mechanical realizations. For $p = 3$, the unitary minimal model

$$\mathcal{M}(3, 4)$$

is the critical Ising model, and the corresponding elliptic double-cover description is governed by the modular lambda function. For $p = 5$, one obtains

$$\mathcal{M}(5, 6), \qquad c = \frac{4}{5},$$

which contains the critical three-state Potts theory as the D-type modular invariant. Thus these two values provide particularly well-known modular realizations within the general family of hypergeometric cyclic covers. In this sense, the corresponding projective monodromy can be related to modular covering constructions, while for generic p the resulting curves are genuinely higher-genus superelliptic curves. The arithmetic properties of the associated triangle groups are governed by the classification of arithmetic triangle groups [9].

1.6 Self-Invertible Primaries

In this subsection, we first identify a universal $\mathbb{Z}_2$-type fusion sector in the unitary minimal models $\mathcal{M}(p, p+1)$. In particular, the primary field $\phi_{p-1,1}$ is self-invertible, with fusion rule $\phi_{p-1,1} \times \phi_{p-1,1} = \mathbf{1}$. This gives an invertible object of order two and hence a natural $\mathbb{Z}_2$ fusion subcategory. Its conformal weight determines whether this invertible object has bosonic or fermionic topological spin, while its Frobenius–Schur indicator provides a further categorical characterization of its self-duality. We will use these structures to relate the fusion and conformal data of the minimal models to the corresponding categorical description.

Claim 1.1. *Unitary minimal model $\mathcal{M}(p, p+1)$ contains an self-invertible primary field a with fusion rule $a \times a = 1$.*

Proof. The identity/vacuum of minimal model denoted by $|h_{1,1}\rangle$. Since the minimal models the fusion rules following $SU(2)$ type factorization

$$|h_{r_1,s_1}\rangle \times |h_{r_2,s_2}\rangle = \sum_{r_3,s_3} |h_{r_3,s_3}\rangle \tag{1.137}$$

where the allowed r_3s and s_3s are

$$r_3 = |r_1 - r_2| + 1, |r_1 - r_2| + 3, \cdots, \min(r_1 + r_2 - 1, 2p - r_1 - r_2 - 1) \tag{1.138}$$
$$s_3 = |s_1 - s_2| + 1, |s_1 - s_2| + 3, \cdots, \min(s_1 + s_2 - 1, 2(p+1) - s_1 - s_2 - 1) \tag{1.139}$$

If we take special primary field $|h_{p-1,1}\rangle$, calculate it self-fusion

$$|h_{p-1,1}\rangle \times |h_{p-1,1}\rangle = |h_{1,1}\rangle \tag{1.140}$$

thus it exist at least one self-invertible primary field in minimal model $\mathcal{M}(p, p+1)$. □

The $\mathbb{Z}_2$ fusion table is

$\times$	$\mathbf{1}$	a
$\mathbf{1}$	$\mathbf{1}$	a
a	a	$\mathbf{1}$

The conformal weight of $|h_{p-1,1}\rangle$ is

$$h_{p-1,1} = \frac{(p-1)(p-2)}{4} \tag{1.141}$$

the corresponding topological spin/conformal spin is

$$\theta_{p-1,1} = e^{2\pi i h_{p-1,1}} = \exp[\frac{\pi i}{2}(p-1)(p-2)] \tag{1.142}$$

we would get

$$\theta_{p-1,1} = \begin{cases} -1, & p = 0, 3 \,(\mathrm{mod}\, 4) \quad \text{fermionic} \\ +1, & p = 1, 2 \,(\mathrm{mod}\, 4) \quad \text{bosonic} \end{cases} \tag{1.143}$$

Recall in boundary Category

Proposition 1.1. *[6] The right Frobenius-Schur indicator satisfies*

$$\nu_X^2 \, dim_R(X) = dim_L(X) \tag{1.144}$$

Similar, the left Frobenius-Schur indicator satisfies

$$dim_L(X) \tilde{\nu}_X^2 = dim_R(X) \tag{1.145}$$

For objects/primary fields of modular categories over $\mathcal{C}$, it is invertible, thus $\nu_X^2 = \tilde{\nu}_X^2 = 1$.

2 Anomaly classification

In QFT/CFT, a central object of interest is the set of conserved quantities, charges, and their associated currents, which determine the rigid symmetries of the theory. When the corresponding Ward identities or conservation laws fail to hold, the theory is said to exhibit an anomaly. In the following subsection, we classify the common types of anomalies that can arise in CFT.

2.1 Algebraic Anomaly

Algebraic anomalies arise naturally in the structure of conformal blocks. For conformal blocks—or, equivalently, spaces of vacua—associated with marked points on a higher-genus surface, such anomalies reflect the nontrivial gluing of conformal blocks defined on different patches. Such algebraic structures and their associated anomalies commonly appear in more general combinatorial two-dimensional models, including, for example, certain constructions of quantum error-correcting codes.

Let Σ be a reduced projective curve, and let

$$P = \{p_1, \dots, p_s\} \subset \Sigma$$

be a collection of distinct smooth marked points. We refer to

$$(\Sigma; p_1, \dots, p_s)$$

as an s-pointed curve. To each marked point p_i, we associate a weight λ_i, regarded as the label of the corresponding representation or module.

The local algebraic structure at the marked points is encoded by a current-type Lie algebra. For $x, y \in \mathfrak{g}$ and functions f, g on Σ, the bracket takes the form

$$[x[f], y[g]] = [x, y][fg],$$

where $x[f]$ denotes the $\mathfrak{g}$-valued current obtained by multiplying x by the function f. After fixing a central element c, the corresponding central extension introduces the central charge seen by a representation. In particular, for a weight λ in the dual space, the central contribution is measured by

$$\lambda_c = (\lambda, c).$$

The combination of the pointed curve, the representations attached to its marked points, and the central extension gives rise to the space of conformal blocks (or space of vacua)

$$\mathcal{V}_\Sigma^\dagger(p_1, \dots, p_s; \lambda_1, \dots, \lambda_s).$$

Thus, a conformal block should not be regarded merely as a vector attached to a fixed algebraic curve: it is a geometric object whose definition depends simultaneously on the complex structure of Σ, the marked points, their weights, and the central extension of the underlying current algebra.

This leads naturally to the question of "transport". As the pointed curve

$$(\Sigma; p_1, \ldots, p_s)$$

moves in a family, the corresponding spaces of conformal blocks form a vector bundle (or, more precisely, a sheaf/vector bundle over the appropriate moduli space) equipped with a natural projectively flat connection. A conformal block can therefore be transported along a path in the moduli of pointed curves. However, this transport need not be strictly compatible with the algebraic structures involved: the central extension and the geometric variation of the curve can produce an additional scalar contribution. We refer to this obstruction to strict algebraic compatibility as the "algebraic anomaly". In this sense, the anomaly is not an extra structure imposed independently of the conformal blocks; rather, it appears when one asks whether the algebraic data defining the conformal blocks can be transported consistently as the underlying pointed curve varies. The resulting failure of strict flatness is typically scalar, so that the natural structure is projectively flat rather than genuinely flat.

Theorem 2.1. *[8] Let $\mathfrak{g}$ be any simple Lie algebra and let $c > 0$ be any central charge. Let (Σ, p) be an irreducible smooth s-pointed curve of any genus $g \geq 0$(where $s > 1$), with weights $(\lambda_1 \cdots \lambda_s)$. Then for space of covacua,*

$$dim\mathcal{V}_\Sigma(p,\lambda) = |T_c|^{g-1} \sum_{\mu \in D_c} \left((\prod_{i=1}^{s} (ch[V(\lambda_i)])) \cdot \prod_{\alpha \in \triangle_+} (2sin(\frac{\pi}{c+h^\vee}\langle \mu+\rho, \alpha\rangle))^{2-2g} \right) \quad (2.1)$$

where D_c is the collection of weight, $\triangle_+$ is the set of positive roots of $\mathfrak{g}$, T

$$|T_c| = (c+h^\vee)^l |P/Q||Q/Q_{lg}| \quad (2.2)$$

, l is the rank of $\mathfrak{g}$, $P(Q)$ is the weight(root) lattice and Q_{lg} is the sublattice of Q generated by the long roots.

For $g = 1, s = 1$ and $\lambda = 0$, $\dim\mathcal{V}_\Sigma(p,\lambda) = |D_c|$.

For the affine scheme of finite type over $\mathbb{C}$,

$$\bar{\Gamma}^{(n)} = \bar{\Gamma}^{(n)}_{\bar{q}} := \mathrm{Spec}\left(\mathbb{C}[Z]/I_G^{\Sigma^*}(n)\right) \quad (2.3)$$

Thus, it could naturally get an ind-affine scheme with filtration

$$\bar{\Gamma}^{(0)} \subset \bar{\Gamma}^{(1)} \subset \cdots \quad (2.4)$$

and

$$\bar{\Gamma}^{(n)}(\mathbb{C}) = \mathrm{Mor}_{\leq n}(\Sigma^*, G) \quad \bar{\Gamma}(\mathbb{C}) = \Gamma = \mathrm{Mor}(\Gamma^*, G) \quad (2.5)$$

there is a bijection,

$$\mathrm{Mor}(\Sigma_R^*, G) \cong \mathrm{Mor}(\mathrm{Spec}\, R, \bar{\Gamma}) \quad (2.6)$$

Under choosing a parabolic subgroup of reductive lie group, these exists an isomorphism of stacks

$$\mathbf{Parbun}_G(\Sigma, P) \cong [\bar{\Gamma} \setminus (\bar{X}_G \times \prod_{i=0}^{s}(G/P_i))] \tag{2.7}$$

$\mathbf{Parbun}_G(\Sigma, P)$ is called quasi-parabolic moduli stack.

For a Riemann surface Σ, the Jacobian is

$$J(\Sigma) = H^0(\Sigma, K)^*/H_1(\Sigma, \mathbb{Z}) \tag{2.8}$$

which is an abelian variety. The generalized theta function generalized as

$$J(\Sigma) \times \mathcal{M}_{g,n} \longrightarrow \mathbb{C} \tag{2.9}$$

The abelian prototype/abelian variety is behind generalized-theta-function. For semistable/stable G-parabolic principle bundle,

$$\mathcal{M}_G(\Sigma_{g,n}) = \mathrm{Bun}_G(\Sigma_{g,n}) \tag{2.10}$$

the k-level bosonic free field relization, the WZW conformal blocks at level k are

$$\mathcal{V}_{\Sigma_{g,n},\lambda}(\mathfrak{g}, k) \cong H^0(\mathcal{M}_G(\Sigma_{g,n}, p), \mathcal{L}^k \otimes \mathcal{E}_\lambda) \tag{2.11}$$

Thus, the conformal blocks are a generalized extension of the classical theta-function construction. For the conformal blocks, the analogous equation of heat equation in ordinary theta function is Knizhnik-Zamolodchikov projective flat connections on bundles.

For finite group H acts on the moduli space, by equivariant Riemann-Roch/Lefschetz theory

$$\mathrm{Tr}(h|H^\bullet(\mathrm{Bun}_G, \mathcal{L})) = \int_{\mathrm{Fix(h)}} \frac{\mathrm{ch}_h(\mathcal{L})\mathrm{Td}(T_{\mathrm{Fix}(h)})}{\det(1 - h|N_{\mathrm{Fix}(h)})} \tag{2.12}$$

If the deformation preserves the fixed-point geometry and the equivariant line bundle data, then the equivariant index does not change.

Remark 1. Minimal models, whether unitary or non-unitary, can be realized through a Coulomb-gas construction with appropriate screening operators. We begin with a free boson equipped with a background charge,

$$T(z) = -\frac{1}{2}(\partial\phi)^2 + i\alpha_0\partial^2\phi. \tag{2.13}$$

For a pair of coprime positive integers (p, q), we introduce the Coulomb-gas parameters [3]

$$\alpha_+ = \sqrt{\frac{q}{p}}, \qquad \alpha_- = -\sqrt{\frac{p}{q}}, \qquad \alpha_+\alpha_- = -1, \qquad \alpha_0 = \frac{\alpha_+ + \alpha_-}{2}. \tag{2.14}$$

The corresponding screening currents are

$$S_+(z) = e^{i\alpha_+\phi(z)}, \qquad S_-(z) = e^{i\alpha_-\phi(z)}, \tag{2.15}$$

whose contour integrals define the screening charges

$$Q_{\pm} = \oint dz, S_{\pm}(z). \tag{2.16}$$

Since the screening currents have conformal weight one, the associated charges commute with the Virasoro algebra,

$$[L_n, Q_{\pm}] = 0, \tag{2.17}$$

or equivalently $[T(z), Q_{\pm}] = 0$ in the contour sense. The degenerate Virasoro primary fields are represented by vertex operators

$$V_{r,s}(z) = e^{i\alpha_{r,s}\phi(z)}, \qquad \alpha_{r,s} = \frac{1-r}{2}\alpha_+ + \frac{1-s}{2}\alpha_-. \tag{2.18}$$

After imposing the appropriate screening/cohomological conditions, this construction yields the Virasoro minimal model $\mathcal{M}(p,q)$ with central charge

$$c = 1 - 24\alpha_0^2 = 1 - \frac{6(p-q)^2}{pq}. \tag{2.19}$$

In particular, the choice $(p,q) = (3,4)$ gives the unitary Ising model $\mathcal{M}(3,4)$ with $c = \frac{1}{2}$, whereas $(p,q) = (2,5)$ gives the non-unitary Lee–Yang model $\mathcal{M}(2,5)$ with $c = -\frac{22}{5}$. Thus, the screening construction applies uniformly to both unitary and non-unitary minimal models; the distinction is encoded in the choice of the Coulomb-gas parameters (p,q).

In terms of the oscillator modes a_n of the free boson, the Virasoro generators take the form [4]

$$L_0 = \frac{1}{2}\sum_{k=0}^{\infty} a_{-k}a_k + \frac{1}{4}a_0^2 - \alpha_0 a_0, \tag{2.20}$$

$$L_n = \frac{1}{4}\sum_{k=1}^{\infty} a_{n-k}a_k - \alpha_0(n+1)a_n. \tag{2.21}$$

The screening charges therefore provide a concrete realization of the minimal-model representation theory inside the free-boson Fock space. For the unitary case, one has an additional and particularly useful realization through the coset construction of $\mathrm{SU}(2)_k$, where the level k is determined by the minimal model under consideration. This description is especially natural in the context of conformal blocks on two-dimensional moduli spaces.

2.2 Bonsonization Anomaly

We next consider the anomaly associated with bosonization. This becomes particularly relevant when realizing a CFT through a nonlinear sigma model. In the fermionic description, the theory naturally decomposes into the four spin-structure sectors$R-R, R-NS, NS-R, NS-NS$ and the corresponding partition functions depend on the choice of spin structure. Upon bosonization, these sectors are reorganized into a bosonic description, and

the transformation between the two descriptions involves a nontrivial Jacobian. The consistency of this Jacobian, together with the transformation of the different spin-structure sectors, provides a natural place for an anomaly to appear.

This construction also makes modularity manifest. On a higher-genus Riemann surface, the fermionic partition functions are naturally expressed in terms of theta functions and spin structures, while the bosonized description is more naturally related to the geometry of the Jacobian variety of the surface. Consequently, the anomaly is not merely a local effect of the field redefinition: it is tied to the global consistency of the bosonization map under changes of spin structure, gluing, and modular transformations. From this perspective, bosonization provides a useful bridge between the local field-theoretic description of the CFT and the global geometry of abelian varieties and their associated theta functions.

Recall ordinary theta function,

$$\theta(z,\tau)=\sum_{n\in\mathbb{Z}}\exp(\pi i n^2\tau+2\pi i n z) \tag{2.22}$$

After bosonization, the fermionic free-field realization of the $\mathcal{N}=1$ theory is naturally expressed in terms of theta functions with characteristics. The four spin structures appearing in the torus partition function, (1.103),(1.104),(1.105),(1.106). For $\vartheta_{0,0}$ and $\vartheta_{0,\frac{1}{2}}$, the leading term of these two theta function is $\sim 1+\cdots$. For $\vartheta_{\frac{1}{2},0}$ and $\vartheta_{\frac{1}{2},\frac{1}{2}}$, the leading term of these two theta function is $\sim q^{1/8}\times\cdots$. The R boundary condition would create half-shift of the bosonic momentum lattice. So $q^{1/8}$ is the vacuum energy shift caused by the half-integer lattice characteristic.

If we write (1.99) in form

$$K_\lambda(\tau)=\frac{1}{\eta(\tau)}\sum_{n\in\mathbb{Z}}q^{\frac{(2pp'n+\lambda)^2}{4pp'}} \tag{2.23}$$

Under the modular transformation $S:\tau\longmapsto-\frac{1}{\tau}$, the theta functions transform as

$$K_\lambda\left(-\frac{1}{\tau}\right)=\sum_{\mu\in\mathbb{Z}_{2pp'}}S_{\lambda\mu}K_\mu(\tau), \tag{2.24}$$

where the S-matrix is

$$S_{\lambda\mu}=\frac{1}{\sqrt{2pp'}}\exp\left(-\frac{2\pi i\lambda\mu}{2pp'}\right)=\frac{1}{\sqrt{2pp'}}\exp\left(-\frac{\pi i\lambda\mu}{pp'}\right) \tag{2.25}$$

Under $T:\tau\longmapsto\tau+1$, the transformation is diagonal,

$$K_\lambda(\tau+1)=\sum_{\mu\in\mathbb{Z}_{2pp'}}T_{\lambda\mu}K_\mu(\tau), \tag{2.26}$$

with

$$T_{\lambda\mu}=\delta_{\lambda\mu}\exp\left[2\pi i\left(\frac{\lambda^2}{4pp'}-\frac{1}{24}\right)\right] \tag{2.27}$$

Clear, the modular data are cyclotomic from previous analysis. If

$$\zeta_N = e^{2\pi i/N}, \qquad N = 2pp' \tag{2.28}$$

then

$$S_{\lambda\mu} = \frac{1}{\sqrt{N}} \zeta_N^{-\lambda\mu}. \tag{2.29}$$

For a Galois automorphism

$$\sigma_\ell : \zeta_N \longmapsto \zeta_N^\ell, \qquad (\ell, N) = 1, \tag{2.30}$$

we obtain

$$\sigma_\ell \left(S_{\lambda\mu} \right) = S_{\ell\lambda,\mu} = S_{\lambda,\ell\mu} \tag{2.31}$$

Thus, the induced Galois action on the theta-function labels is

$$\lambda \longmapsto \ell\lambda \pmod{2pp'} \tag{2.32}$$

In particular,

$$\sigma_\ell : \vartheta_{\frac{2pp'-1}{2pp'},0} \longmapsto \vartheta_{\frac{\ell(2pp'-1)}{2pp'},0} = \vartheta_{-\frac{\ell}{2pp'},0} \tag{2.33}$$

The minimal-model are obtained by selecting the appropriate Galois-compatible combinations of theta functions $\vartheta_{\frac{1}{2},0}$. Hence the passage from the free-field theta-function realization to the rational minimal model can be viewed as a restriction of the cyclotomic theta-function representation together with its Galois action. Since the free fermion, the Ramond sector shifts the momentum lattice by 1/2, producing the characteristic vacuum factor $q^{1/8}$. Therefore, the $q^{1/8}$ factor could be understood as the elementary bosonization manifestation of the same shifted-lattice phenomenon whose rational, while the redundancy of extension of theta function gives cyclotomic organization which is encoded by the Galois action on the extended set theta functions.

2.3 Trace Anomaly

For a conformal field theory defined on a two-dimensional Riemannian manifold $(M, g_{\mu\nu})$, classical Weyl invariance implies the Ward identity

$$\langle T^\mu{}_\mu \rangle = 0. \tag{2.34}$$

However, after quantization, this identity is generally violated due to the regularization of the functional determinant. Using the heat-kernel expansion of the Laplace-type operator determined by the metric $g_{\mu\nu}$, one finds the two-dimensional trace anomaly

$$\langle T^\mu{}_\mu \rangle = -\frac{c}{24\pi} R \tag{2.35}$$

where c is the central charge and R is the scalar curvature of M. This anomalous trace represents a quantum violation of the classical Weyl Ward identity and is therefore referred to as the Weyl/trace anomaly.

For instance, a sphere

$$\int_{S^2} \sqrt{g} R = 4\pi\chi(S^2) = 8\pi \tag{2.36}$$

thus

$$\int_{S^2} \sqrt{g} \langle T^\mu_\mu \rangle = \frac{c}{24\pi} \times 8\pi = \frac{c}{3} \tag{2.37}$$

For our case, two layer system, 4D

$$\langle T^\mu_\mu \rangle = \frac{1}{16\pi^2}(cW_{\mu\nu\rho\sigma}W^{\mu\nu\rho\sigma} - aE_4 + b\Box R) \tag{2.38}$$

where

$$E_4 = R_{\mu\nu\rho\sigma}R^{\mu\nu\rho\sigma} - 4R_{\mu\nu}R^{\mu\nu} + R^2 \tag{2.39}$$

For $S^2 \times S^2$

$$E_4 = 2R_1R_2 \tag{2.40}$$

thus

$$\int_{S^2\times S^2} E_4 dV = 2(\int_{S^2} R_1 dV_1)(\int_{S^2} R_2 dV_2) = 128\pi^2 \tag{2.41}$$

2.4 Coset Anomaly

The coset anomaly arises when one constructs a conformal field theory associated with a coset G/H starting from a Wess–Zumino–Witten (WZW) model based on a Lie group G. In the coset construction, the degrees of freedom associated with the subgroup $H \subset G$ are gauged and subsequently removed through a BRST quantization and gauge-fixing of H. Although the classical gauged WZW action is invariant under the local H gauge symmetry, the quantum path-integral measure and the determinants generated by gauge fixing and ghost fields can produce nontrivial anomalous contributions. In the BRST formulation, this issue is reflected in the requirement that the BRST operator remain nilpotent at the quantum level.

For

$$G = \frac{\prod_{i=1}^n G_i}{H} \tag{2.42}$$

where each G_i is the connected semisimple lie group, H is the subgroup of the center of $\prod_{i=1}^n G_i$, thus

$$\dim H \leq \sum_i^n \operatorname{rank} G_i \tag{2.43}$$

For WZW model with a gauge field A, which A is an imaginary one-form,[7]

$$A = A^{10} + A^{01}, \quad A^{10} = -(A^{01})^* \tag{2.44}$$

A^{10} is dz part of A, while A^{01} is $d\bar{z}$ part of A. The coupling part would be

$$\frac{i}{2\pi}\int_\Gamma [\langle A^{10}_R, g^{-1}\bar{\partial}g\rangle_k + \langle g\partial g^{-1}, A^{01}_L\rangle_k + \langle gA^{10}_R g^{-1}, A^{01}_L\rangle_k - \langle A^{10}, A^{01}\rangle_{\tilde{k}}] \tag{2.45}$$

the notation $\tilde{k}$ denote the length of the long roots of the root systems. For different length of root system, here the innner product need rescale. Thus, here we denote $\tilde{k}$. L

and R denote the left moving realization and right moving realization of the field A. For canonical embedding, case L equal R, which symmetric case(corresponding the spherical reps Lie groups). For group G level k

$$[J_n(x), J_m(y)] = J_{n+m}([x,y]) + \langle x,y\rangle_k n\delta_{n+m,0} \tag{2.46}$$

the central charge is

$$c_{G,k} = \sum_i \frac{k_i \dim G_i}{k_i + g_i^\vee} = \frac{k \dim G}{k + g^\vee} \tag{2.47}$$

and for $H^C/H \times H$, where H_C is the complexification of H,it is a noncompact case since here consider the left-right invariant functional space $L^2(H^C)$ usually not compact, take

$$[J_n, J_m] = J_{m+n}([x,y]) - \langle x,y\rangle_{\bar{k}} n\delta_{n+m,0} \tag{2.48}$$

thus

$$c_{H^C/H,\bar{k}} = \sum_j \frac{\bar{k}_j \dim H_J}{\bar{k}_j - h_j^\vee} = \frac{\bar{k} \dim H}{\bar{k} - h^\vee} \tag{2.49}$$

in total

$$c_{G/H} = c_{G,k} + c_{H^C,H,\bar{k}} + c_{\text{ghost}} \tag{2.50}$$

where

$$c_{\text{ghost}} = -2\dim H \tag{2.51}$$

it comes from the free commuting ghosts in the adjoint representation of H^C.

2.5 Chiral Anomaly

In this section, we discuss chiral anomalies from two complementary perspectives. In the conventional field-theoretic treatment, the chiral anomaly is closely related to the chiral structure of fermions and, in particular, to the properties of the chirality operator γ_5 in the Clifford algebra. Although the classical theory may possess a conserved chiral current, this conservation law can fail after quantization, with the corresponding Ward identity acquiring an anomalous contribution. This phenomenon is usually understood through the non-invariance of the fermionic path-integral measure, or equivalently through the Jacobian associated with a chiral transformation. In this sense, the Clifford-algebraic structure of the fermions provides the local field-theoretic origin of the anomaly.

Once the theory is formulated in terms of conformal blocks, however, there is another natural mechanism through which chiral anomaly-like behavior can arise. In a two-dimensional CFT, the conformal blocks encode the global and algebraic structure of the theory, including the fusion and braiding of primary fields. In particular, when a self-invertible primary field a, satisfying $a \times a = \mathbf{1}$, has a nontrivial exchange phase with itself, the corresponding conformal blocks can acquire a sign or phase under the exchange of identical insertions. This provides a second realization of chiral behavior, now expressed directly in the algebraic structure of the conformal blocks rather than through the explicit γ_5 construction. We therefore separate the discussion into two subsections: the first treats the conventional γ_5- and fermion-based chiral anomaly, while the second studies the corresponding phenomenon from the perspective of conformal blocks, self-invertible primary fields, and their braiding structure.

2.5.1 Chiral Anomaly caused by γ_5

In two dimensional cases

$$\{\gamma^\mu, \gamma^\nu\} = 2g^{\mu\nu} \tag{2.52}$$

the γ^5 is defined by

$$\gamma^5 = \gamma^0\gamma^1 \tag{2.53}$$

which satisfied

$$\{\gamma^5, \gamma^\mu\} = 0 \quad (\gamma^5)^2 = 1 \tag{2.54}$$

for massless case,

$$S = \int d^2x\bar{\psi}i\gamma^\mu D_\mu\psi \tag{2.55}$$

for axial transformation

$$\psi \to e^{i\alpha\gamma^5}\psi \tag{2.56}$$

the current corresponding to this transformation is

$$j_5^\mu = \bar{\psi}\gamma^\mu\gamma^5\psi \tag{2.57}$$

for classical case, the current conservation/Ward identity

$$\partial_\mu j_5^\mu = 0 \tag{2.58}$$

also by Heat kernel method could give the violation of Ward identity

$$\partial_\mu j_5^\mu = \frac{q}{2\pi}\epsilon^{\mu\nu}F_{\mu\nu} \tag{2.59}$$

The Dirac spinor could decompose into two chiral components in light-cone coordinates,

$$\psi = \psi_L + \psi_R \quad \gamma^5\psi_R = \psi_R \quad \gamma^5\psi_L = -\psi_L \tag{2.60}$$

in these two chiral components, (2.59) becomes

$$\partial_- j_R = +\frac{q}{4\pi}\epsilon^{\mu\nu}F_{\mu\nu} \quad \partial_+ j_L = -\frac{q}{4\pi}\epsilon^{\mu\nu}F_{\mu\nu} \tag{2.61}$$

Thus, the chiral anomaly by gamma matrix reflect the imbalance between left- and right-moving quantum currents.

2.5.2 Chiral Anomaly caused by twisted Hecke Actions

For a two-dimensional conformal field theory, one may begin with a target space $\Gamma_{g,n}$ on which the conformal blocks are well defined. When the theory admits a realization in terms of generalized theta functions associated with the Jacobian of the target curve, this construction naturally leads to a moduli space. In this geometric setting, the framework of Geometric Langlands can arise naturally. It is important, however, to keep in mind that this description is based on a semisimple Lie algebra $\mathfrak{g}$, or equivalently a semisimple Lie group G. The corresponding theory is the one before introducing screening operators;

once screening charges are included, one is led instead to Coulomb-gas realizations and, in particular, to minimal models.

Under this framework, to each rank-n holomorphic vector bundle E on a complex algebraic curve X, equipped with a suitable holomorphic connection, one can associate a Hecke eigensheaf on the moduli space $\mathrm{Bun}_n(X)$ of rank-n holomorphic vector bundles on X. The Hecke operators then provide the geometric counterpart of the eigenvalue structure, relating the geometry of the moduli space to the chiral data of the conformal theory.

In the abelian case, one can consider analogous structures arising from algebraic varieties, in a way reminiscent of the construction of error-correcting codes from algebraic geometry, including topological-code constructions such as the toric code. In such a setting, line defects can carry nontrivial topological information.

For example, consider a $\mathbb{Z}_2$ line defect e satisfying

$$e \times e = 1.$$

Although this fusion rule is fixed, the corresponding defect category admits two possible associativity structures, distinguished by

$$H^3(\mathbb{Z}_2, U(1)) \cong \mathbb{Z}_2.$$

For the trivial class one may choose

$$F_e^{eee} = +1,$$

whereas for the nontrivial Dijkgraaf–Witten class the associator can be chosen as

$$F_e^{eee} = -1.$$

when

$$H^3(\mathbb{Z}_2, U(1)) \cong \mathbb{Z}_2 \tag{2.62}$$

and

$$\omega(a,b,c) = \begin{cases} -1, & (a,b,c) = (1,1,1) \\ +1, & \text{otherwise} \end{cases} \tag{2.63}$$

Hence the Dijkgraaf–Witten cocycle does not change the fusion rule $e \times e = 1$, but changes the associativity data of the defect operators. When such a defect sector is coupled to a chiral CFT, this additional phase can enter the consistency conditions for gauging the symmetry and is naturally interpreted in terms of the corresponding chiral anomaly.

Now we write this into formal Hecke actions. Let X be a smooth projective complex curve and G a reductive group. The natural geometric object is the moduli stack

$$\mathrm{Bun}_G(X)$$

of G-bundles on X. For a representation $V \in \mathrm{Rep}(G^\vee)$, the Hecke correspondence at a point $x \in X$ is described by

$$\mathrm{Bun}_G(X) \xleftarrow{q} \mathrm{Hecke}_{V,x} \xrightarrow{p} \mathrm{Bun}_G(X),$$

and defines a Hecke functor

$$\mathsf{H}_{V,x}(\mathcal{F}) = q_!\left(p^*\mathcal{F} \otimes \mathcal{K}_V\right),$$

where $\mathcal{K}_V$ is the kernel associated with V through geometric Satake.

For a Hecke eigensheaf $\mathcal{F}_E$ associated with a $G^\vee$-local system E on X, one has schematically

$$\mathsf{H}_{V,x}(\mathcal{F}_E) \simeq V_E|_x \otimes \mathcal{F}_E.$$

The proposed geometric identification is that the defect action is realized by a Hecke correspondence,

$$\mathsf{D}_e \simeq \mathsf{H}_e$$

Thus, the same cocycle appears in the composition of Hecke operators:

$$(\mathsf{H}_e\mathsf{H}_e)\mathsf{H}_e = -\mathsf{H}_e(\mathsf{H}_e\mathsf{H}_e)$$

Thus, $\mathbb{Z}_2$-twisted defect sector can be realized geometrically through Hecke correspondences, with the nontrivial H^3-class appearing as the associativity obstruction of the resulting action on chiral conformal blocks. This construction naturally extends to the moduli space equipped with a spin, or equivalently metaplectic, structure. In this setting, the fermionic defect a is represented by a Hecke-type correspondence $T^a_{p^2}$ acting on the corresponding chiral conformal blocks or determinant line. The fermionic self-statistics, $R_{aa} = -1$, then implies that the interchange of two such defect insertions is accompanied by the fermionic braiding phase. Schematically, one may write

$$T^a_{p^2} \circ T^a_{q^2} = R_{aa}, T^a_{q^2} \circ T^a_{p^2}, \qquad R_{aa} = -1, \tag{2.64}$$

provided the Hecke correspondences are understood as the corresponding fermionic, spin-lifted defect operators rather than ordinary commuting arithmetic Hecke operators.

2.6 Pre-screening Hecke effects on primary fields

We now consider the spectral structure of pre-screening minimal model, which is $SU(2)_k$ WZW to provide a naturel initial data for an arithmetic Hecke random walk. Our purpose is not to identify the Hecke evolution with some real dynamical process, but rather to construct a nature scale/quantitatively accessible observable: the decay of arithmetic Fourier modes under repeated Hecke averaging. The staring point WZW model supplies a distinguished spectral distribution through its Virasoro and quantum KdV structure, while the arithmetic side which the Hecke operators and Hecke eigenvalues governing the subsequent evolution.

Here we consider affine $SU(2)_k$ since after a well-choosen screening operator, it would lead to minimal model. For this model, the central charge is

$$c_k = \frac{3k}{k+2} \tag{2.65}$$

and the affine current

$$J^a(z) = \sum_{n\in\mathbb{Z}} J^a_n z^{-n-1} \quad a = 1, 2, 3 \tag{2.66}$$

with relations

$$[J^a_m, J^b_n] = ic^{ab}{}_c J^c_{m+n} + \frac{k}{2} m\, \delta^{ab}\delta_{m+n,0} \tag{2.67}$$

the stress tensor is obtained by the Sugawara construction

$$T(z) = \frac{1}{k+2} : J^a J^a : (z) \tag{2.68}$$

the affine theory naturally contains a Virasoro algebra on which the quantum KdV hierarchy can be defined.

The integrable structure is described by a family of mutually commuting quantum KdV charges

$$I_1, I_3, I_5, \dots \tag{2.69}$$

The first charge is

$$I_1 = L_0 - \frac{c_k}{24}$$

while the first nontrivial higher charge may be written, in a standard normalization, as

$$I_3 = 2\sum_{n=1}^{\infty} L_{-n}L_n + L_0^2 - \frac{c_k+2}{12}L_0 + \frac{c_k(5c_k+22)}{2880}$$

The precise normalization of the higher charges will not be important below; what matters is that they are simultaneously diagonalizable and therefore provide genuine spectral observables of the pre-screening theory.

The primary field are labeled by representations with conformal weights

$$j = 0, \frac{1}{2}, 1, \dots, \frac{k}{2} \quad h_j = \frac{j(j+1)}{k+2} \tag{2.70}$$

A general state is in form

$$J^{a_1}_{-n_1} \cdots J^{a_m}_{-n_m}|j\rangle$$

and its zero mode is determined by

$$L_0 = h_j + N \qquad N = \sum_{i=1}^{m} n_i.$$

Consequently,

$$I_1(j, N) = h_j + N - \frac{c_k}{24}$$

For a theory placed on a circle of circumference R, the physical energy associated with I_1 is

$$E_\Psi = \frac{2\pi}{R}\mathcal{I}_1(\Psi)$$

Thus the dimensionless ground state energy data can be converted into physical spectral scales only after the macroscopic scale R has been introduced. This is analogous to the arithmetic construction, where the dual lattice variable ℓ^* is dimensionless and acquires an

interpretation as a frequency only after the underlying geometric normalization has been specified.

Let

$$\chi_\ell(x) = e^{2\pi i\langle \ell, x\rangle}$$

be the character we introduced. (It is nature on torus.) For Hecke eigenstates

$$H\chi_\ell = \lambda_\ell \chi_\ell \quad H^N\chi_\ell = \lambda_\ell^N \chi_\ell \tag{2.71}$$

thus, the associated spectral damping rate is therefore

$$\Gamma_\ell = -\log|\lambda_\ell|.$$

Under the extended Riemann hypothesis, the Hecke eigenvalue is bounded schematically by[1]

$$|\lambda_\ell| \lesssim \frac{[\mathrm{Pic}^0_{K,\mathfrak{m}} : G]\log B}{\sqrt{B}} \log(B^n|\Delta_K|N(\mathfrak{m})q_\infty(\chi_\ell))\,,$$

where the archimedean conductor grows with the dual arithmetic frequency,

$$q_\infty(\chi_\ell) \lesssim \left(4 + \frac{4\pi}{\sqrt{n}}\|\ell^*\|\right)^n.$$

Thus the arithmetic estimate gives a relation of the form

$$\|\ell^*\| \longrightarrow q_\infty(\chi_\ell) \longrightarrow |\lambda_\ell| \longrightarrow \Gamma_\ell$$

For comparing the conformal spectrum with the arithmetic Fourier spectrum, we may introduce a localized spatial profile associated with the primary. We approximate it by a Gaussian,

$$\phi_j(x) = A_j \exp\left(-\frac{x^2}{2\sigma_j^2}\right). \tag{2.72}$$

Its Fourier transform is

$$\widehat{\phi}_j(q) = A_j\sqrt{2\pi}, \sigma_j \exp\left(-\frac{\sigma_j^2 q^2}{2}\right). \tag{2.73}$$

Therefore the parameter σ_j determines the range of Fourier modes populated by the conformal probe. A simple leading estimate can be obtained by using the first KdV charge as the spectral scale. We define

$$q_j^{(1)} = \frac{2\pi}{R}\sqrt{\mathcal{I}_1(j)}, \tag{2.74}$$

and therefore estimate the characteristic width by

$$\sigma_j \sim \frac{1}{q_j^{(1)}} = \frac{R}{2\pi\sqrt{\mathcal{I}_1(j)}}. \tag{2.75}$$

If the vacuum Casimir contribution is omitted when defining the characteristic excitation scale, one may instead use h_j and write

$$\sigma_j \sim \frac{R}{2\pi\sqrt{h_j}} = \frac{R}{2\pi}\sqrt{\frac{k+2}{j(j+1)}}. \tag{2.76}$$

This last expression should be regarded as a spectral estimate rather than an exact statement about the spatial profile of a primary field.

After N Hecke steps, the resulting observable can therefore be written schematically as

$$\mathcal{A}_j(N) = \sum_{\ell^*} \widehat{\phi}_j(\ell^*)\lambda_\ell'^{N}. \tag{2.77}$$

in detail

$$\mathcal{A}_j(N) = \sum_{\ell^*} \exp\left(-\frac{\sigma_j^2 q_\ell'^2}{2}\right)\lambda_\ell'^{N} \tag{2.78}$$

The effective decay rate associated with a given conformal probe can consequently be defined by

$$\Gamma_j^{\mathrm{eff}} = -\lim_{N\to\infty}\frac{1}{N}\log\left|\frac{\mathcal{A}_j(N)}{\mathcal{A}_j(0)}\right|. \tag{2.79}$$

This quantity is the natural observable to compare across the $SU(2)_k$ primary spectrum.

3 Global Double-Layer Geometry

3.1 The three-dimensional gluing geometry

We now show that the double-layer geometry satisfies a systolic-geometric condition. The relevant spatial manifold is taken to be

$$M = S^2 \times S^1, \tag{3.1}$$

where the S^2 factor describes the spatial geometry of each layer, while the S^1 direction parametrizes the gluing of the two layers.

Consider the product metric

$$ds^2 = R_2^2, ds_{S^2}^2 + R_1^2, d\theta^2, \qquad \theta \sim \theta + 2\pi. \tag{3.2}$$

The relevant stable systoles are represented by the two factors of the product. In particular,

$$\mathrm{stsys}_2(M) = 4\pi R_2^2, \qquad \mathrm{stsys}_1(M) = 2\pi R_1, \tag{3.3}$$

while

$$\mathrm{Vol}(M) = (4\pi R_2^2)(2\pi R_1). \tag{3.4}$$

Consequently,

$$\frac{\mathrm{stsys}_2(M)\,\mathrm{stsys}_1(M)}{\mathrm{Vol}(M)} = 1. \tag{3.5}$$

Thus, for the product metric, the double-layer geometry has a finite and scale-independent systolic ratio. More generally, Gromov's stable systolic inequality for products of spheres gives

$$\frac{\mathrm{stsys}_2(M)\,\mathrm{stsys}_1(M)}{\mathrm{Vol}(M)} \leq C, \qquad M = S^2 \times S^1, \tag{3.6}$$

where C is a universal constant independent of the particular Riemannian metric. Therefore, the double-layer construction is compatible with the systolic geometry of $S^2 \times S^1$. In this sense, the corresponding geometric constraint is not an additional dynamical assumption of the CFT, but follows from the topology and metric geometry of the underlying spatial manifold.

There is, however, an additional geometric condition in the present construction. The systolic geometry is required to admit a hyperelliptic realization. A compact Riemann surface Σ_g of genus g is called hyperelliptic if it admits a conformal involution

$$\iota : \Sigma_g \longrightarrow \Sigma_g, \qquad \iota^2 = 1, \tag{3.7}$$

such that the quotient is the Riemann sphere,

$$\Sigma_g/\langle\iota\rangle \cong \mathbb{P}^1. \tag{3.8}$$

Equivalently, Σ_g admits a degree-two holomorphic map to $\mathbb{P}^1$. The involution has precisely $2g+2$ fixed points, which are the Weierstrass points, and the surface can be represented as

$$y^2 = \prod_{i=1}^{2g+2} (x - x_i). \tag{3.9}$$

We first clarify the role of the curve appearing in our conformal-block construction. The curve

$$C_p : \qquad y^N = x^{p(p+2)}(1-x)^p(1-zx)^{-(p+1)}, \qquad N = p(p+1), \tag{3.10}$$

should not itself be identified with the hyperelliptic double-layer surface. It is an N-cyclic superelliptic cover and arises as the Euler-period realization of the hypergeometric conformal blocks.

Remark 2. It is useful to distinguish three different appearances of the sphere in the construction. First, the BPZ equation is formulated on the four-punctured projective line $\mathbb{P}^1_z$, whose marked points are represented by $\{0, \mathrm{z}, 1, \infty\}$. *Thecorrespondingprojectivemonodromydescribestheg*

Second, the Euler-period construction introduces the cyclic superelliptic curve C_p. Its canonical order-two subgroup gives a degree-two quotient, and the resulting quotient is $C_p/\mathbb{Z}_2 \simeq \mathbb{P}^1$. This is the spherical quotient associated with the cyclic covering construction.

Third, the global gluing geometry considered below contains a spherical base in the circle bundle $S^1 \hookrightarrow M \longrightarrow S^2, \qquad M \simeq S^2 \times S^1$. The latter S^2 is used as the base of the three-dimensional global geometry and should not be identified with either of the previous spheres without specifying an additional map.

Thus, the relevant structures are related but play different roles:

$$\begin{aligned} \text{BPZ/projective geometry} &\longrightarrow \mathbb{P}^1_z, \\ \text{cyclic period curve} &\longrightarrow C_p/\mathbb{Z}_2 \simeq \mathbb{P}^1, \\ \text{global gluing geometry} &\longrightarrow S^1 \hookrightarrow S^2 \times S^1 \longrightarrow S^2. \end{aligned}$$

The common four-point data provide the natural link between these constructions, but the corresponding spheres should be regarded as distinct geometric realizations unless an explicit identification is constructed.

For the genus-one case, the four distinguished points of the four-point conformal block are $\{0, 1, z^{-1}, \infty\}$. The corresponding double cover of the sphere can be written, up to a Möbius transformation and an overall rescaling, as

$$X_1^{(2)}: \qquad w^2 = x(x-1)(1-zx). \tag{3.11}$$

Thus, in the genus-one case, there is a direct correspondence

$$\boxed{\text{four-point hypergeometric geometry}} \qquad \Longleftrightarrow \qquad \boxed{\text{genus-one } \mathbb{Z}_2 \text{ surface.}} \tag{3.12}$$

For higher genus, we retain the same four distinguished points and add the remaining branch points required by the hyperelliptic double cover. We therefore consider

$$X_g^{(2)}: \quad w^2 = x(x-1)(1-zx)\prod_{j=1}^{2g-2}(x-a_j), \tag{3.13}$$

where the a_j provide the additional hyperelliptic moduli. The four points $\{0, 1, z^{-1}, \infty\}$ are thus embedded into the $2g+2$ branch points of the hyperelliptic surface.

Both C_p and $X_g^{(2)}$ are naturally mapped to the same x-sphere,

$$C_p \xrightarrow{\pi_p} \mathbb{P}^1_x, \qquad \pi_p(x, y) = x, \tag{3.14}$$

and

$$X_g^{(2)} \xrightarrow{\pi_2} \mathbb{P}^1_x. \tag{3.15}$$

This allows us to define their fiber product

$$\mathcal{C}_{p,g} = C_p \times_{\mathbb{P}^1} X_g^{(2)}. \tag{3.16}$$

The point of this construction is that C_p and $X_g^{(2)}$ need not be the same curve. Rather, they retain different pieces of the same conformal-block geometry: C_p encodes the Euler-period realization of the hypergeometric block, while $X_g^{(2)}$ encodes the hyperelliptic double-layer geometry.

Correspondingly, the four-point conformal block can be extended to the hyperelliptic geometry by inserting the additional branching operators,

$$\left\langle \phi_1(0)\phi_2(z)\phi_3(1)\phi_4(\infty)\prod_{j=1}^{2g-2} V(a_j) \right\rangle_{\mathbb{P}^1}. \tag{3.17}$$

The four original insertions determine the original conformal-block problem, while the additional $V(a_j)$ encode the extra branch points of the double-layer surface.

We can make the relation between the cyclic curve C_p and the $\mathbb{Z}_2$ geometry more explicit. The deck group of C_p is

$$G = \mathbb{Z}_N, \tag{3.18}$$

with $N = p(p+1)$. Since N is always even, G contains a unique order-two subgroup. Equivalently, we may define $w = y^{N/2}$. The resulting degree-two quotient satisfies

$$w^2 = x^{\alpha_0}(1-x)^{\alpha_1}(1-zx)^{\alpha_z}, \tag{3.19}$$

where

$$\alpha_0 = p(p+2), \qquad \alpha_1 = p, \qquad \alpha_z = p+1. \tag{3.20}$$

The exponent at infinity is determined by the divisor relation,

$$\alpha_\infty = -\left(\alpha_0 + \alpha_1 + \alpha_z\right), \tag{3.21}$$

so that, up to an irrelevant sign,

$$|\alpha_\infty| = p^2 + 2p - 1. \tag{3.22}$$

Since the quotient is a $\mathbb{Z}_2$ cover, only the parity of these local exponents is relevant. We therefore obtain two cases. For odd p,

$$\alpha_0 \equiv \alpha_1 \equiv 1 \pmod 2, \qquad \alpha_z \equiv \alpha_\infty \equiv 0 \pmod 2, \tag{3.23}$$

so the points $\{0, 1\}$ are the two branch points of the resulting degree-two cover. For even p,

$$\alpha_0 \equiv \alpha_1 \equiv 0 \pmod 2, \qquad \alpha_z \equiv \alpha_\infty \equiv 1 \pmod 2, \tag{3.24}$$

and instead $\{z^{-1}, \infty\}$ are the two branch points.

Thus, in both parity cases, the canonical order-two quotient of C_p has exactly two branch points. We can determine the topology of the quotient directly from the Riemann–Hurwitz formula. For a degree-d holomorphic map

$$f : \Sigma_g \longrightarrow \Sigma_{g'},$$

the formula relates the genera of the two surfaces by

$$2g - 2 = d(2g' - 2) + B, \tag{3.25}$$

where B is the total ramification contribution. In the present case, the order-two quotient is a degree-two map

$$C_p/\mathbb{Z}_2 \longrightarrow \mathbb{P}^1,$$

so $d = 2$.

Moreover, a simple branch point of a double cover has ramification index two and hence contributes one to B. Since the $\mathbb{Z}_2$ quotient has precisely two branch points, the total ramification contribution is $B = 1 + 1 = 2$. Therefore,

$$2g - 2 = 2(-2) + 2 = -2, \tag{3.26}$$

and hence

$$g = 0. \tag{3.27}$$

Thus the canonical $\mathbb{Z}_2$ quotient of C_p does not produce a nontrivial higher-genus hyperelliptic surface. Its quotient has genus zero and is therefore conformally equivalent to the sphere,

$$\boxed{C_p/\mathbb{Z}_2 \cong \mathbb{P}^1 \cong S^2.} \tag{3.28}$$

In this sense, after reducing the cyclic Euler-period geometry to its canonical $\mathbb{Z}_2$ sector, the resulting double-layer geometry has the topology of S^2, with the two surviving branch points specifying the corresponding two-sheeted covering structure.

Therefore the natural quotient of the Euler-period curve is not itself the desired higher-genus hyperelliptic surface, but rather

$$C_p \longrightarrow H_p \cong \mathbb{P}^1. \tag{3.29}$$

This observation is important for the interpretation of the construction. The N-cyclic Euler-period curve does contain a canonical $\mathbb{Z}_2$ sector, but this sector alone supplies only two branch points. It therefore gives a genus-zero quotient rather than the full hyperelliptic double-layer geometry.

The two parity sectors are equivalent under Möbius transformations. For odd p, for example, one may use

$$M_{odd}(x) = \frac{x}{1-x} : 0 \mapsto 0 \quad 1 \mapsto \infty \tag{3.30}$$

then the Z_2 branch pair becomes $\{0, \infty\}$, For even p, an affine Möbius transformation can take

$$M_{even}(x) = x - z^{-1} : z^{-1} \mapsto 0 \quad \infty \mapsto \infty \tag{3.31}$$

Thus the two cases are equivalent at the level of the resulting degree-two covering geometry.

To obtain the hyperelliptic surface required by the double-layer construction, we therefore retain the two branch points selected by the canonical $\mathbb{Z}_2$ quotient and supplement them by the remaining $2g$ branch points. Schematically, for odd p, this gives

$$X^{(2)}_{g,p} : \quad w^2 = x(1-x)\prod_{j=1}^{2g}(x - a_j), \tag{3.32}$$

while for even p the corresponding construction is obtained from the pair

$$\{z^{-1}, \infty\}$$

after a Möbius transformation. In either case, the resulting surface has $2g+2$ branch points and therefore defines a genus-g hyperelliptic double cover

$$X^{(2)}_{g,p} \longrightarrow \mathbb{P}^1. \tag{3.33}$$

In this way, the two constructions should not be identified directly. The curve C_p is the Euler-period realization of the hypergeometric conformal block, while $X^{(2)}_g$ is the hyperelliptic double-layer geometry. Their relation is mediated by the common four-punctured sphere, the induced $\mathbb{Z}_2$ sector, and, more generally, the fiber product

$$\mathcal{C}_{p,g} = C_p \times_{\mathbb{P}^1} X^{(2)}_g. \tag{3.34}$$

The hyperelliptic structure therefore does not require the Euler-period curve itself to be hyperelliptic. Rather, the Euler-period construction supplies the conformal-block data on the four-punctured sphere, while the double-layer construction extends this geometry to the required Z_2-branched hyperelliptic surface.

Since here we consider the two-layer description. The hyperelliptic involution

$$\iota : X^{(2)}_g \longrightarrow X^{(2)}_g, \qquad \iota^2 = 1,$$

exchanges the two sheets of the double cover

$$X^{(2)}_g \longrightarrow \mathbb{P}^1 \simeq S^2.$$

Thus, the S^2 appearing here is the base geometry obtained after quotienting by the layer-exchange involution. To describe the gluing between the two layers, it is natural to introduce a continuous circle direction. In this sense, the S^1 factor can be regarded as the geometric realization of the gluing direction between the two layers, while the corresponding discrete layer exchange is encoded by the Z_2 involution.

This leads naturally to the three-dimensional manifold

$$S^1 \hookrightarrow E \xrightarrow{\pi} S^2.$$

Oriented circle bundles over S^2 are classified by their first Chern class

$$c_1(E) \in H^2(S^2;\mathbb{Z}) \cong \mathbb{Z}.$$

For $c_1(E) = 0$, the circle bundle is trivial,

$$E \cong S^2 \times S^1,$$

and consequently admits a global section. For $c_1(E) \neq 0$, the bundle is generally non-trivial and there is no global section. The standard example is the Hopf fibration

$$S^1 \hookrightarrow S^3 \longrightarrow S^2,$$

whose first Chern class is $c_1 = 1$. In this case the circle fiber can only be trivialized locally. More generally, an oriented circle bundle over S^2 with Chern class k has total space

$$E_k \simeq L(k,1),$$

with the $k = 0$ case understood as

$$L(0,1) \simeq S^2 \times S^1.$$

For the trivial bundle relevant to the present double-layer construction, we therefore take

$$M = S^2 \times S^1.$$

The S^2 factor is the quotient geometry of the two layers, while the S^1 factor keeps track of the gluing direction between them. This gives a direct geometric realization of the double-layer picture as a three-dimensional manifold.

The relevant homology groups are

$$H_1(M;\mathbb{Z}_2) = H_1(S^2;\mathbb{Z}_2) \oplus H_1(S^1;\mathbb{Z}_2) \cong \mathbb{Z}_2,$$

generated by the circle $[S^1]$. On the other hand,

$$H_2(M;\mathbb{Z}_2) = H_2(S^2;\mathbb{Z}_2) \oplus H_2(S^1;\mathbb{Z}_2) \cong \mathbb{Z}_2,$$

generated by $[S^2 \times \{\mathrm{pt}\}]$.

Thus the two non-trivial homology classes relevant for the systolic problem are represented by the 1-cycle S^1 and the 2-cycle $S^2 \times \{\mathrm{pt}\}$. Their corresponding $\mathbb{Z}_2$-systoles are

$$\mathrm{stsys}_1^{\mathbb{Z}_2}(M), \qquad \mathrm{stsys}_2^{\mathbb{Z}_2}(M).$$

In the higher-genus construction, the same picture is understood as the double-layer geometry together with the additional hyperelliptic branch data. The quotient by the layer-exchange Z_2 involution remains the sphere, while the non-trivial genus is carried by the branched double cover

$$X_g^{(2)} \longrightarrow S^2.$$

The corresponding three-dimensional geometry may then be studied through the product or circle-fibration structure over this quotient geometry. In particular, the relevant systolic ratio takes the form [5]

$$\frac{\mathrm{stsys}_3^{\mathbb{Z}_2}(M)}{\mathrm{stsys}_2^{\mathbb{Z}_2}(M)\ \mathrm{stsys}_1^{\mathbb{Z}_2}(M)} \leq \frac{g}{g\sqrt{\log g}} = \frac{1}{\sqrt{\log g}} \longrightarrow 0, \qquad g \longrightarrow \infty.$$

Thus the large-genus limit suppresses the three-dimensional systolic scale relative to the product of the 1- and 2-dimensional systolic scales. This provides the desired connection between the hyperelliptic double-layer construction and the systolic-geometric regime.

3.2 $S^2 \times S^2$ Weyl Anomaly and Modular Diffusion

We regard

$$\mathcal{P} = S^2 \times S^2 \tag{3.35}$$

as a pre-target manifold for the two-CFT system. A map from $\mathcal{P}$ to the two CFT layers associates the two S^2 factors with the two layers,

$$S^2 \times S^2 \longrightarrow \mathcal{C}_1 \oplus \mathcal{C}_2. \tag{3.36}$$

The relevant point is that this construction admits a stable section in the corresponding geometric realization. Thus, the stability of the double-layer configuration is associated with the existence of this stable section, rather than being inferred merely from the curvature of $S^2 \times S^2$.

This distinction is important because, in general, Ising-type CFT configurations, including those with curvature or anomaly contributions of opposite sign, can undergo a phase transition to the toric-code (TC) phase. Therefore, the spherical geometry alone does not guarantee stability. In the present construction, the stable section provides the additional geometric structure that distinguishes the stable double-layer realization from such potentially unstable deformations.

We next consider the geometric realization of the double-layer construction. The two layers are related by a $\mathbb{Z}_2$ involution

$$\iota : X_g^{(2)} \longrightarrow X_g^{(2)}, \qquad \iota^2 = 1, \tag{3.37}$$

whose quotient gives the spherical geometry

$$X_g^{(2)}/\langle \iota \rangle \simeq \mathbb{P}^1 \simeq S^2. \tag{3.38}$$

Thus, the S^2 should be understood as the quotient geometry of the two-layer system, rather than as one of the two layers themselves. The involution exchanges the two sheets, while the branch points encode the loci at which the two sheets are connected.

For the double-layer realization considered here, the first quasiparticle correlation function agrees with its corresponding spherical realization. We take this agreement as a direct CFT-level check that the double-layer construction preserves the relevant spherical sector. In particular, the introduction of the second layer does not modify the leading quasiparticle data characterizing the spherical realization.

For the two Ising layers,

$$c_1 = c_2 = \frac{1}{2}, \qquad c_{\mathrm{dbl}} = c_1 + c_2 = 1. \tag{3.39}$$

Their trace-anomaly contributions therefore have the same sign and add rather than cancel. Schematically,

$$\langle T^{\mu}{}_{\mu} \rangle_{\mathrm{dbl}} \propto (c_1 + c_2) R, \tag{3.40}$$

up to the overall sign convention used for the trace anomaly. Hence the two-layer construction is compatible with a spherical positive-curvature realization[2]: the two identical Ising sectors reinforce the curvature response rather than producing a cancellation.

The curvature/anomaly agreement should be understood as a consistency check rather than as the origin of the stability statement. In particular, positive curvature by itself does not imply CFT stability. Here the stable section provides the geometric stability condition, while the agreement of the quasiparticle correlation functions gives a direct CFT-level check. The curvature and anomaly analysis then supplies an independent geometric consistency check for the same double-layer realization.

The geometry also naturally separates two structures that should not be identified. The discrete exchange of the two layers is described by $\mathbb{Z}_2$, whereas the continuous direction

associated with gluing the two layers is represented by a circle. We therefore consider the trivial circle bundle

$$S^1 \hookrightarrow M \longrightarrow S^2, \qquad M \simeq S^2 \times S^1. \tag{3.41}$$

Here the S^2 describes the spherical quotient of the double-layer geometry, while the S^1 provides the continuous direction along which the two layers are glued. Thus,

$$\boxed{\text{layer exchange} = \mathbb{Z}_2, \qquad \text{inter-layer gluing direction} = S^1.} \tag{3.42}$$

The two structures consequently play complementary roles: $\mathbb{Z}_2$ acts discretely on the two layers, whereas S^1 supplies the continuous geometric direction of the gluing.

For a genus-one modular parameter, the natural target space is the modular moduli space

$$\mathcal{M}_{1,1} = \mathbb{H}/SL(2,\mathbb{Z}),$$

with the usual identification

$$\tau_\alpha \mapsto \frac{a\tau_\alpha + b}{c\tau_\alpha + d}, \qquad \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in SL(2,\mathbb{Z}).$$

The modular parameter should therefore be regarded as a point in the quotient space rather than as a particular representative in the upper half-plane.

The S^1 direction allows these modular data to be continuously transported between the two layers. Introducing a coordinate

$$s \in S^1,$$

we describe the modular data of each quasiparticle sector by

$$\tau_\alpha = \tau_\alpha(s).$$

The complete modular configuration is consequently a map

$$S^1 \longrightarrow \prod_\alpha \mathcal{M}_{1,1}, \qquad s \longmapsto \{\tau_\alpha(s)\}_\alpha.$$

This formulation is important because the double layer remains a fixed geometric object: the continuous deformation occurs in the modular data carried by the individual sectors, rather than in the number or identity of the layers.

A natural metric on the upper half-plane is the Poincaré metric,

$$ds^2_{\mathbb{H}} = \frac{d\tau\, d\bar{\tau}}{(\operatorname{Im}\tau)^2}.$$

The corresponding modular-invariant distance is obtained after quotienting by $SL(2,\mathbb{Z})$,

$$d_{\mathcal{M}}([\tau],[\tau']) = \inf_{\gamma \in SL(2,\mathbb{Z})} d_{\mathbb{H}}(\tau, \gamma\cdot\tau'),$$

where

$$d_{\mathbb{H}}(\tau,\tau') = \operatorname{arcosh}\left(1 + \frac{|\tau-\tau'|^2}{2\,\operatorname{Im}\tau\,\operatorname{Im}\tau'}\right).$$

Thus modular-equivalent representatives have zero distance in $\mathcal{M}_{1,1}$, as required.

The corresponding diffusion of modular parameters along the gluing circle can be described by a natural geometric diffusion functional

$$S_{\text{diff}} = \sum_{\alpha} \kappa_\alpha \int_{S^1} ds\, \frac{|\partial_s \tau_\alpha|^2}{(\operatorname{Im}\tau_\alpha)^2},$$

with κ_α parametrizing the stiffness of the corresponding quasiparticle sector. More invariantly, this is the energy of the map $S^1 \longrightarrow \prod_\alpha \mathcal{M}_{1,1}$.

The construction therefore permits a continuous diffusion of modular data without changing the underlying double-layer topology or its $\mathbb{Z}_2$ identification.

For example, two representatives

$$\tau_1 = i, \qquad \tau_2 = 1 + i$$

describe the same point of $\mathcal{M}_{1,1}$ because

$$\tau_2 = T \cdot \tau_1, \qquad T : \tau \mapsto \tau + 1.$$

Their modular distance is consequently zero. In contrast,

$$\tau_1 = i, \qquad \tau_2 = 2i$$

represent genuinely different points, with

$$d_{\mathbb{H}}(i, 2i) = \operatorname{arcosh}\frac{5}{4} \simeq 0.693.$$

The latter therefore gives a genuine modular displacement along the gluing direction.

The resulting picture is that of a fixed, spherical double-layer CFT together with sector-dependent modular data that can diffuse continuously along the S^1 direction. The $\mathbb{Z}_2$ structure controls the exchange of the two layers, the spherical S^2 describes their quotient geometry, and the S^1 supplies the continuous direction along which the modular data are transported. In this sense, modular diffusion is a deformation of the internal CFT data over a fixed double-layer geometry, rather than a deformation that introduces additional layers or assigns one universal modular parameter to each layer.

4 Discussion and Outlook

In this work, we start from the BPZ equations of unitary minimal models and follow the geometry of their conformal blocks. The hypergeometric form of the BPZ equation gives

the projective monodromy, while the Euler-period representation gives the cyclic curve C_p. The important point is that the order-two subgroup of the cyclic deck group gives

$$C_p/\mathbb{Z}_2 \simeq \mathbb{P}^1 \simeq S^2. \tag{4.1}$$

This gives a natural spherical geometry associated with the conformal blocks. We then use this sphere as the quotient geometry of a hyperelliptic double-layer construction. In this picture, the $\mathbb{Z}_2$ involution exchanges the two layers, while the S^1 direction describes the continuous gluing between them, giving the global geometry

$$S^1 \hookrightarrow M \longrightarrow S^2, \qquad M \simeq S^2 \times S^1. \tag{4.2}$$

It is important that the different curves and quotients in this construction are not identified with each other. The cyclic curve C_p comes from the period representation of the conformal blocks, while the hyperelliptic curve describes the double-layer geometry. Their relation is through the common spherical quotient. Similarly, the $\mathbb{Z}_2$ layer exchange and the S^1 gluing direction describe two different structures. This distinction is useful when we introduce the modular data: the double-layer geometry is kept fixed, while different quasiparticle sectors can have different modular parameters τ_α. The S^1 direction then gives a natural direction along which these modular parameters can vary. We proposed the corresponding diffusion functional as a geometric description of this variation.

For the stability question, we use $S^2 \times S^2$ as a pre-target manifold for the two-CFT system. The map to the two layers admits a stable section, and this is the point we use to distinguish the stable double-layer realization from a general spherical configuration. In particular, positive curvature by itself is not enough to establish stability, since Ising-type CFT configurations can in general be driven through a phase transition to the toric-code phase. The stable section is therefore the relevant geometric ingredient for the stability statement.

The Weyl anomaly gives another, independent check of the construction. For the two Ising layers,

$$c_1 = c_2 = \frac{1}{2}, \qquad c_{\mathrm{dbl}} = 1, \tag{4.3}$$

so the two anomaly contributions have the same sign and add rather than cancel. This is compatible with the positive-curvature realization on $S^2 \times S^2$. Together with the agreement of the relevant quasiparticle correlation function with the spherical realization, this gives a consistent CFT interpretation of the double-layer geometry. We therefore distinguish the roles of these observations: the stable section is related to stability, while the correlation functions and Weyl anomaly provide consistency checks.

There are several directions that we would like to understand further. In particular, it would be useful to make the relation between the stable section and the moduli of the hyperelliptic double layer more explicit. It would also be interesting to understand whether the modular diffusion introduced here can be obtained directly from the CFT rather than being introduced as a geometric functional. Finally, the higher-genus systolic constructions suggest that the same picture may be extended beyond the simplest $S^2 \times S^1$

geometry. In this sense, the present construction provides a starting point for relating conformal-block monodromy, algebraic curves, double-layer geometry, and modular data in a common geometric framework.

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